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Second cycle, 30 credits

Designing a Carbonator Focused on Interaction Design

Enhancing User Experience through Electrical Integration

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Abstract

This degree project in Industrial Design Engineering was conducted at KTH Royal Institute of Technology in collaboration with Aarke, a Swedish company renowned for its premium kitchen appliances. The goal was to improve the interaction design of the Aarke Carbonator Pro, a carbonation device for glass bottles, by addressing usability challenges related to intuitiveness and the overall user experience.

Following the Double Diamond methodology, the project included extensive user research, product study, competitor analysis, and interviews. Key findings highlighted difficulties with the locking mechanism, unclear user feedback, and a lack of intuitive flow during the carbonation process. Based on these insights, a new project direction was defined: to explore how electric functionality could be integrated into the product to enhance interaction, reduce complexity, and improve the overall user experience.

The focus was specifically on the user interaction during the carbonation sequence, without altering the internal carbonation mechanics or gas infusion system. Several concepts were developed and evaluated through an iterative prototyping process. The outcome includes a high-fidelity CAD model and a full-scale prototype demonstrating a new interaction flow supported by electronic features while maintaining Aarke's brand values of refinement, minimalism, and timeless design.

This project highlights the value of user-centered design in the development of premium products. It demonstrates how refined interaction design can enhance usability, emotional engagement, and brand alignment. The resulting concept is recommended as a foundation for future development of the Carbonator Pro.

Keywords

Interaction Design, User Experience, Emotional Design, Carbonator, Kitchen Appliances, Human-Machine Interaction, Industrial Design, Usability, Product Development

Sammanfattning

Detta examensarbete inom Teknisk Design har genomförts vid Kungliga Tekniska högskolan i samarbete med Aarke, ett svenskt företag som specialiserar sig på premiumprodukter för köket. Projektets syfte var att förbättra interaktionsdesignen för Aarke Carbonator Pro, en kolsyremaskin avsedd för glasflaskor, med fokus på att öka användarvänligheten och den övergripande användarupplevelsen.

Med utgångspunkt i Double Diamond-modellen genomfördes en omfattande designprocess bestående av användarstudier, produktanalys, konkurrentjämförelser och intervjuer. Analysen identifierade brister i produktens intuitivitet, särskilt kopplat till låsmekanismen, otillräcklig återkoppling till användaren samt bristande flöde i användarinteraktionen vid kolsyrning.

Dessa insikter låg till grund för en omformulering av projektets inriktning, där fokus lades på att undersöka hur elektriska funktioner kan integreras för att förenkla användningen, förstärka användarinteraktionen och förbättra upplevelsen utan att förändra den befintliga kolsyremekaniken.

Arbetet koncentrerades på användarens interaktion under kolsyrningsförloppet och omfattade framtagning och utvärdering av flera koncept genom iterativ prototyputveckling. Det slutgiltiga resultatet består av en högupplöst CAD-modell samt en fullskalig prototyp som demonstrerar ett nytt interaktionsflöde med elektroniskt stöd, samtidigt som Aarkes kärnvärden bevaras.

Examensarbetet tydliggör vikten av användarcentrerad design vid utveckling av premiumprodukter och visar hur genomtänkt interaktionsdesign kan bidra till ökad användbarhet, emotionellt engagemang och varumärkesmässig samstämmighet. Det framtagna konceptet föreslås som en grund för fortsatt utveckling av Carbonator Pro.

Nyckelord

Interaktionsdesign, Användarupplevelse, Emotionell design, Kolsyremaskin, Köksprodukter, Människa-maskin-interaktion, Industridesign, Användbarhet, Produktutveckling

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Chapter 1

Introduction

This section introduces the degree project, providing an overview of the project. It includes a background, followed by a problem description. The purpose and objectives will be outlined, along with the scope and limitations of the project. Additionally, the methods used in the project will be described.

1.1 Background

"For over a decade, Aarke has been a Swedish pioneer in the design of premium products for the home that transform everyday routines into meaningful rituals. We call it everyday engineering." aarke (n.d.-c).

Aarke, established in 2013 by industrial designers Carl Ljungh and Jonas Groth, is a Swedish design company dedicated to creating premium kitchen appliances that seamlessly blend functionality with aesthetics (aarke, n.d.-c). The two designers saw a lack of stylish Italian-inspired carbonators on the market and decided to capitalize on this opportunity by creating a product to fill that niche. From there on, the company Aarke AB has developed into a brand which offers not only carbonators but coffee makers, kettles, and purifiers (aarke, n.d.-a). Driven by a deep understanding of human behavior, Aarke focuses on re-imagining overlooked aspects of everyday design, bringing thoughtful innovation to the heart of the home (aarke, n.d.-c).

The Aarke flagship product and the foundation of the company's success, is their stainless steel carbonator. They currently offer two models: Carbonator 3 and Carbonator Pro, see Figure 1.1 (aarke, n.d.-a). The Carbonator 3 carbonates water using a plastic bottle and represents the third iteration of this product, with each version refined and improved. The Carbonator Pro, on the other hand, carbonates

water in a glass bottle and still remains in its first iteration (aarke, n.d.-a).



(a) (b)

Figure 1.1: Aarke Carbonators: (a) Carbonator 3 and (b) Carbonator Pro aarke, n.d.-a.

While the Carbonator 3 features a sleek and user-friendly design, the Carbonator Pro adopts a more robust and complex structure. This is due to the use of glass bottles, which require improved safety measures during the carbonation process. Unlike plastic, glass has significantly lower elasticity, leading to more severe consequences if it breaks under high pressure. To prevent this, the Carbonator Pro includes additional reinforcements to reduce the risk of bottle breakage (Groth, Co-Founder of Aarke, personal communication, 2025).

Carbonators, as a product category, lack established norms for user interaction. Aarke's two carbonator models exemplify this by employing different interactions, further highlighting the absence of a standardized design. This lack of consistency makes it challenging to create an intuitive, user-friendly product. However, it also presents a unique opportunity to explore innovative solutions and redefine user engagement, ultimately resulting in a more streamlined and optimized user experience.

1.2 Problem Description

While the Carbonator Pro boasts an exceptional design, it has faced challenges and received criticism, as users find its operation difficult to understand. As mentioned above, carbonating a glass bottle requires additional measures to ensure a secure procedure, resulting in a more complex design, compared to Carbonator 3. Furthermore, the Carbonator Pro is still in its first iteration, and as a relatively new design, interactions issues are therefore not uncommon.

1.3 Purpose and Objectives

The purpose of this degree project is to develop an improved interaction design for Aarke's Carbonator Pro, with the goal of enhancing the overall user experience and addressing current usability challenges. The focus lies on creating a more intuitive, safe, and enjoyable interaction that aligns with Aarke's brand identity and design philosophy.

The objectives of the project are:

1. Improve Intuitiveness: *Redesign the interaction flow so that the product is self-explanatory, with affordances and visual cues that clearly communicate how the carbonator is to be used.*
2. Enhance Safety Perception: *Develop interaction mechanisms that not only ensure user safety during carbonation with glass bottles, but also convey a feeling of security and robustness throughout the process.*
3. Streamline User Flow: *Minimize the number of interaction steps required to complete the carbonation process by combining functions and optimizing their placement and resistance, to support an efficient and fluid experience.*
4. Clear Feedback: *Ensure that the product provides immediate, understandable, and consistent feedback, helping users recognize system responses and feel confident in their actions.*
5. Respect Brand Identity: *Align the interaction and aesthetic decisions with Aarke's values of timelessness, minimalism, and refinement, ensuring the new concept fits within their existing product portfolio.*

The project outcomes include a detailed concept developed through iterative prototyping, presented as a high-fidelity CAD model along with proposals for

internal mechanics. Additionally, a physical prototype is created to represent the scale, form, and user interaction, though not functionally operative.

1.4 Scope and Limitations

The scope of this project involves improving the user experience of Carbonator Pro by addressing key usability challenges. This encompasses the process of securing the glass bottle, as well as the interaction involved in carbonating the water. Additional interactions may be incorporated if the user study identifies them as necessary in order to improve the overall functionality of the product. The insertion of the CO_2 cylinder will be taken in to account, but will not be a focus area of this project.

During this project, only the Human-Machine Interaction (HMI) will be considered. The carbonation process itself, including mechanical or technical aspects of gas infusion as well as the quality of the sparkling water, is beyond the scope of this study and will therefore be disregarded. Furthermore, the project will not involve any thorough work regarding the physical hardware of the carbonator, as the materials and manufacturing process is intended to remain the same. Additionally, the project will not cover aspects related to the long-term durability or performance of the product.

As this is a conceptual design project, the final product will not include detailed specifications for specific components. Instead, feasibility will be demonstrated by referencing comparable products or suggesting potential components that could fulfill the intended functions. The CAD model is intended to communicate form, interaction, and overall structure, and therefore is not optimized for material use or designed with precise features to secure internal components in place.

1.5 Methods Description

Throughout the project, the Double Diamond model was utilized to facilitate an iterative and structured design process (Design Council, n.d.). This framework consists of two sequential diamonds, each comprising two stages: an initial phase of divergent thinking, where a broad range of ideas and possibilities are explored, followed by a convergent phase, where solutions are refined and focused. A visual representation of the model is provided in Figure 1.2.

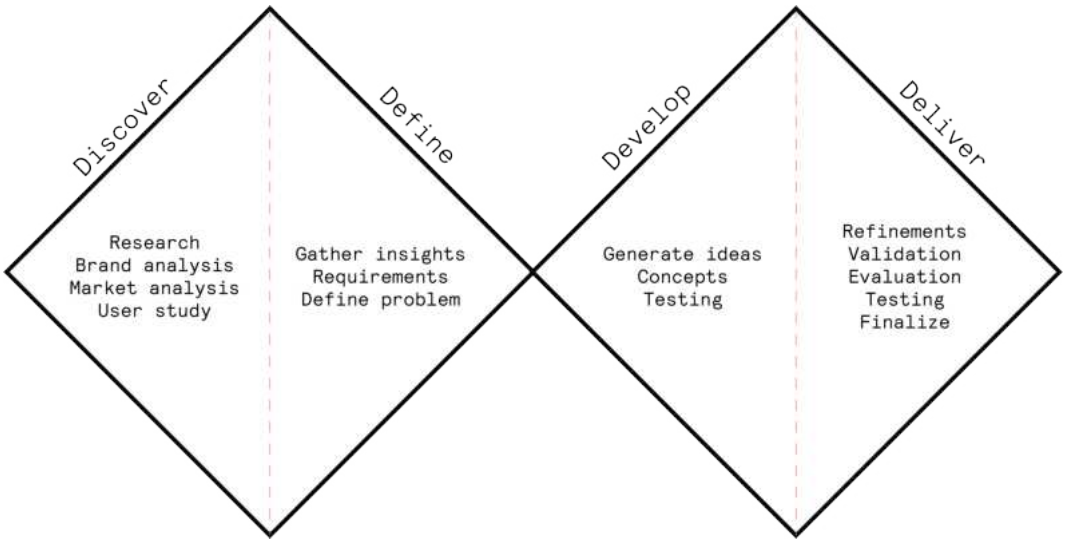


Figure 1.2: The Double Diamond Model

The project was structured into the four phases described in the Double Diamond model. Each phase encompassing a variety of methods and tasks.

Phase 1: Discover
The initial phase of the project focused on exploring the problem space and gaining a broad understanding of the context. This included a background study of Aarke as a company and its current products, as well as a literature review covering key topics such as interaction design, emotional design, and the technical functioning of carbonators.

In addition, various user-centered methods were applied, including observations, interviews, and practical testing of current products. The goal was to uncover patterns in user behavior, identify sources of confusion or dissatisfaction, and explore what constitutes a positive experience. Insights from the user studies were complemented by an analysis of the product's design and functionality, including a technical teardown of the existing carbonator model.

This phase aimed to uncover user pain points, evaluate existing solutions, and build a solid knowledge foundation for the following stages.

Phase 2: Define
In the Define phase, findings from the Discover phase were synthesized to identify key patterns, user needs, and usability challenges. The goal was to narrow the focus of the project by translating broad insights into concrete design opportunities.

Methods such as clustering and requirements formulation were used to structure and prioritize the information gathered from user studies, interviews, and technical analysis. Special attention was given to identifying which aspects of interaction caused the most confusion or friction for users, and which areas offered the greatest potential for improvement.

This phase was primarily analytical, aiming to frame a clear design challenge based on real user behavior and expectations. It culminated in the development of a project definition, which set the foundation for the subsequent design and concept development phases.

Phase 3: Develop
During the Develop phase, the focus shifted toward exploring potential solutions based on the challenges and opportunities identified in the previous stages. Building on the insights and defined requirements, various concepts were generated and iteratively developed through creative and experimental design work.

Ideas and concepts were iterated through brainstorming, low-fidelity sketches and mockups, and some early testing. The goal of this phase was to converge on a design direction that demonstrated both functional improvements and a more intuitive user experience. A chosen concept could then further be developed in the last project phase.

Phase 4: Deliver
In the final phase, the selected concept was further developed and refined using CAD and interaction-focused prototyping tools. Focus was on developing a finalized design and solution. The design was validated and compared to the initial product.

Chapter 2

Frame of Reference

A frame of reference was conducted to gather background information regarding interaction design. Several design principles were observed to gain a deep understanding about the topic. Additionally emotional design och interaction design specifically in home appliances were investigated.

2.1 Interaction Design

Interactive design, or designing for human interaction, is a broad field that encompasses principles, guidelines, and methods that aim to create artifacts that resonate with and are understood by people (Kurniawan, 2004). This study investigates the concept of interaction design and examines how it can contribute to the process of designing effective and user-friendly products. During the literature search for this frame of reference it was observed that interaction design is often closely associated with graphic design and digital interfaces. However, this study will not focus on that aspect. Instead, the emphasis will be on interaction design in relation to physical products. Nevertheless, insights from digital design may also inform the discussion where relevant.

Given the broad scope of this subject (Kurniawan, 2004), the following areas have been selected for in-depth exploration: Design Principles for Interaction, Emotional Design, and Interaction Design in Home Appliances. These areas were chosen for their significant impact on creating user-friendly, intuitive, and emotionally engaging physical products. Each section will examine how these concepts enhance user interaction and contribute to effective product design, as well as how they will be incorporated into this project.

Design Principles for Interaction Design

When discussing various design principles for interaction, four frameworks will be introduced: Don Norman's six principles, Jakob Nielsen's five usability components as well as his ten heuristics, and Miles A. Kimball's 38 design principles. Together, these theories provide a comprehensive foundation for understanding the principles of interaction design.

Don Norman's six principles
The first contribution comes from Don Norman, the author of *The Design of Everyday Things*, where he examines the principles behind designing everyday objects such as doors and stoves. In his work, he introduces six fundamental design principles that emphasize usability and user-centered design, offering valuable insights that are highly applicable to interaction design of physical products (Norman, 1988).

The first principle, *Visibility*, emphasizes that the more visible an element is, the more likely users are to notice and understand how to use it. Norman (1988) argues that designers must therefore carefully consider what the user should notice and interact with first. Conversely, if an element is difficult to see, it becomes harder for users to engage with it effectively.

The second principle, *Feedback*, ensures that users are aware of the actions they have performed and the results of those actions. This can be communicated through various means, such as visual, auditory, or tactile signals. According to Norman (1988), if users are left guessing about the outcome of their actions, the product will be perceived as confusing and difficult to use. Clear and predictable interactions not only enhance usability but also build trust in the product and the brand.

When discussing feedback, many people might immediately think of digital feedback, such as a display showing battery usage or an indicator light turning on when a screen is activated. However, feedback is equally important in mechanical tasks. Take a vacuum cleaner as an example. If a vacuum cleaner were completely silent while in use, users might wonder whether it's functioning at all. However, hearing a faint sound of debris being sucked up provides reassurance. This auditory feedback confirms that the action was successful, enhancing the overall user experience (Vieira et al., 2017).

Norman's third principle, *Constraints*, highlights the importance of limiting the possibilities for interaction to simplify a product's usability. If a product offers too many points of interaction, it can become unclear in what ways or order they should be used, leading to confusion. However, this does not mean the design must be plain

or dull. Instead, Norman (1988) emphasizes the need to focus the content and design on the essentials, ensuring clarity and ease of use without compromising creativity or functionality.

The next principle, *Mapping*, focuses on creating a clear relationship between controls and the effects they have on the world. Norman (1988) uses the example of a stove top: by positioning the controls in the same pattern as the burners, it becomes more intuitive to understand which control corresponds to which burner. This is much easier than if the controls were arranged in a single line, where the mapping would be less clear and potentially confusing. This principle was also disused by Kaley (2018) where she states

> ”*When a design's controls follow real-world conventions and correspond to desired outcomes (called natural mapping), it's easier for users to learn and remember how the interface works. This helps to build an experience that feels intuitive.*” - Kaley (2018).

She continues by arguing that leveraging familiarity with the real world goes beyond just visual resemblance; cultures often have established metaphors and actions that must be considered. For example, in Western cultures, people tend to associate ”more” with ”up” and ”less” with ”down.” Therefore, when designing a product like a speaker, an upward arrow should represent a higher volume, aligning with these cultural associations to make the interaction more intuitive (Kaley, 2018).

The fifth principle, *Consistency*, refers to the importance of using similar operations for similar tasks. Not only does this make the product easier to use, but it also allows users to learn how to operate it more quickly, as the interface contains familiar actions. This principle can be applied not only to a single product but also across an entire product line or even within a broader market category (Norman, 1988).

The last principle is about *Affordance* which describes how the attribute of an object allows people to know how to use it. As an example, Norman (1988) uses a computer mouse, where the physical button gives the cue that it can be clicked to perform an action. Objects with strong affordances are easy to understand how to use because their design inherently suggests the intended interaction.

The research done for the frame of reference has shown that the concept of affordance is widely used in design and has been discussed extensively, not only in Norman's work but also in various other studies. You and Chen (2007) explore affordances from an industrial design perspective in their article *Applications of Affordance and Semantics in Product Design*. According to them, affordance refers to a complex three-way relationship between users, objects, and actions. It represents a product's

inherent potential to enable user actions without requiring memory, reasoning, or interpretation. By interacting with the products, users can initiate a series of potential actions to ultimately achieve their goals (You and Chen, 2007). They state that

"*Affordance is about action but not communication*" - You and Chen (2007).

Continuing, affordances can be categorized in several ways. First, they can be classified based on their polarity, whether they are positive or negative, depending on the user's perspective (Parmentier et al., 2021). Parmentier et al. (2021) uses the example of a safety gate which provides positive affordance for adults by allowing access while providing negative affordance for children by preventing access. Second, affordances can be divided based on the perceptual information available about them, which results in four types: perceptible affordances (visible and actionable), hidden affordances (actionable but not visible), false affordances (seemingly actionable but not functional), and correct rejections (neither actionable nor misleading). Perceptible and hidden affordances enable actions, while false affordances and correct rejections do not (Parmentier et al., 2021).

Another classification considers spatial and temporal relationships. Affordances can be nested, where they are spatially connected, or sequential, where they are revealed over time through interaction. For example, a door handle demonstrates both nested and sequential affordances (Parmentier et al., 2021).

Lastly, Parmentier et al. (2021) argues that affordances also can be classified based on interacting systems: artefact-artefact affordances (AAA), which describe interactions between product components, and artefact-user affordances (AUA), which refer to interactions between the user and the product. Considering both types of affordances is particularly important for modular or reconfigurable products to ensure their functionality and usability.

By understanding and applying these classifications, designers can better align product features with user actions and interpretations, thereby creating more intuitive and meaningful interactions (Parmentier et al., 2021).

Nielsen's five usability components and ten heuristics
Usability can be a concept that is difficult to define but as stated by ISO 9241-11:208 usability is

"*Extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use*" - International Organization for Standardization (2018).

Nielsen (2012) describes usability as a quality that assesses how easy it is for users to interact with a user interface. It can be defined through five key components:

1. *Learnability*: How easy it is for users to use the product for the first time.
2. *Efficiency*: How quickly users can perform tasks once they have learned how the product works.
3. *Memorability*: How easy it is for users to remember how to perform tasks after a significant period of not using the product.
4. *Errors*: How many errors users make, how severe they are, and how easily they can be corrected.
5. *Satisfaction*: How pleasant and enjoyable it is for users to interact with the product.

Nielsen (1994) also introduces a set of design principles known as heuristics, which function as general guidelines rather than rigid rules, intended to assist designers in creating user-friendly products. Several of these heuristics, including feedback, consistency, constraints, and mapping, align with four of Norman's principles. Additionally, Nielsen outlines six more heuristics, three of which are particularly relevant to physical products: user control and freedom, error prevention, and help and documentation.

User control and freedom imply that users should have the ability to easily undo or redo actions, giving them the flexibility to explore without fear of making irreversible mistakes. It ensures that users feel in charge of their interactions with the system, allowing them to correct errors quickly and confidently. This principle reduces frustration, making the experience more efficient (Nielsen, 1994).

Continuing with error prevention, good design goes beyond providing error messages and actively working to prevent issues before they occur. By eliminating error-prone conditions or checking for potential mistakes before an action is finalized, users are given the opportunity to confirm their intentions, reducing the risk of errors. Two types of errors are mentioned, slips and mistakes. The difference is that slips are unconscious and mistakes conscious where the user's expectations or understanding of the system do not align with the design (Nielsen, 1994).

Lastly, help and documentation should be available when the users need additional support. Ideally, the system should be intuitive enough that users don't require extra guidance. However, when documentation is necessary, it should be concise and focused on the tasks to resolve issues efficiently (Nielsen, 1994).

Miles A. Kimballs's 38 Design Principles
Kimball (2013) has through a literature study compiled 38 design principles. Two of these principles, emphasis and proximity, are relevant for interaction with physical products and not mentioned by the two previous authors.

Emphasis underscores the importance of directing attention to key elements in a design. By highlighting these focal points, users can more easily understand what to prioritize and how to interpret the design. This can be achieved through techniques like contrast, strategic positioning, and separating important elements from others. These methods help guide the user's focus, ensuring that the design communicates its purpose clearly and supports efficient interaction (Carter, 2003; Lidwell et al., 2003).

Proximity, on the other hand, emphasizes the idea that elements placed close together are often perceived as related. By carefully adjusting the distance between elements, designers can create clear groupings and relationships without the need for borders or visual separators. This approach helps in organizing the design more effectively, allowing users to intuitively understand how different elements are connected. It also contributes to a cleaner, more streamlined aesthetic, enhancing both usability and visual appeal (Lidwell et al., 2003).

Don Norman's six principles, Jakob Nielsen's five usability components with his ten heuristics, and Miles A. Kimball's 38 design principles represent three distinct frameworks for design and usability, each serving a different purpose and focus. Nielsen's components primarily provide a set of criteria for assessing user experience and interaction effectiveness, acting as a practical checklist to ensure the design is user-friendly. Norman's principles delve deeper into the specifics of interaction design, offering practical advice for creating products that minimize usability issues and enhance user understanding. Kimball's principles, meanwhile, offer a broader, more comprehensive approach to design, incorporating elements such as visual organization and product grouping that complement the more interaction-focused advice from Norman and Nielsen. When combined, these frameworks provide a well-rounded foundation for creating intuitive, functional, and visually cohesive products that meet the needs of users.

Throughout the project, this information will be valuable in several phases. During the research phase, it will help identify and understand the current issues with the machine. During observations, it will provide insight into why users interact with the machine the way they do. It will also serve as a benchmark for evaluating the new iteration to determine if it offers improvements. Finally, in the design phase,

it will guide the development of a well-thought-out and user-friendly design for the next iteration.

Emotional Design

Emotional design plays a central role in shaping user experience by fostering an emotional connection between the user and the product. While functionality enables users to achieve their goals efficiently, emotional design elevates the interaction, contributing to satisfaction, engagement, and long-term attachment. Don Norman identifies three cognitive levels of emotional design: visceral, behavioral, and reflective. These levels must be considered together to create meaningful and pleasurable user experiences (Norman, 2004).

At the visceral level, users respond instinctively to the product's sensory qualities, for instance its form, color, materials, and overall aesthetic appeal. This level is driven by immediate perception and helps shape the user's first impression. Desmet (2002) expands on this concept by demonstrating how specific design elements can evoke targeted emotional responses. In his emotion-driven design framework, emotions are triggered through visual cues, symbolic meaning, and the interaction between product and user. For instance, clean lines, premium materials, and elegant finishes can convey feelings of trust, calm, or excitement (Desmet, 2002).

The behavioral level focuses on usability and the quality of interaction. Here, the product must enable users to carry out tasks effortlessly, providing a sense of control and reliability. Marc Hassenzahl introduces a useful distinction between pragmatic and hedonic qualities of a product. Pragmatic quality relates to functionality and effectiveness, while hedonic quality refers to stimulation, self-expression, and emotional fulfillment. Thus, while behavioral design ensures task success, it also supports deeper experiential needs when it considers users' personal values and enjoyment (Hassenzahl, 2005).

At the reflective level, emotional responses are shaped by the user's thoughts and evaluations after the interaction. This includes long-term associations with the product and brand, such as a sense of pride, environmental consciousness, or lifestyle alignment (Norman, 2004). Hassenzahl (2005) further emphasizes that users often reflect on their experiences through a lens of meaning and identity, and that product experiences become significant when they contribute to a user's self-concept or narrative.

Together, these three levels form a holistic framework for emotional design. Desmet (2002) argues that emotion is not a byproduct of design but a design

goal in itself, and that designing for specific emotions can enhance the overall experience. Emotional design not only increases satisfaction, but also strengthens the relationship between the user, the product, and the brand (Desmet, 2002; Norman, 2004). Importantly, these three levels, visceral, behavioral, and reflective, are not isolated but dynamically interlinked, reinforcing one another across the full user journey (Hassenzahl, 2005; Norman, 2004).

In the context of this project, where the objective is to design a carbonator for Aarke, a company known for its focus on aesthetic quality and user-centered design, these principles are highly relevant. At the visceral level, the carbonator should immediately communicate quality and desirability through its materiality and form. At the behavioral level, the interaction should be smooth, intuitive, and efficient, offering users a sense of control and flow. Finally, at the reflective level, the product should align with broader values such as sustainability, innovation, or timeless craftsmanship, strengthening long-term emotional attachment (Desmet, 2002; Hassenzahl, 2005; Norman, 2004).

Interaction Design in Home Appliances

Kitchen and home appliances are integral to our daily lives, simplifying routine tasks and enhancing convenience. These products often feature a variety of interaction interfaces, such as control panels, handles, knobs, and buttons, all designed to facilitate seamless user interaction (Macedo and Silva, 2014). Interaction design focuses on optimizing these interfaces to ensure they are intuitive, efficient, and satisfying to use. A well-executed interaction design can significantly improve the overall user experience (UX), which encompasses all aspects of a user's interaction with a product. This includes not only direct interaction with the physical elements of the appliance but also the emotional and cognitive responses, such as ease of use, reliability, and satisfaction derived from the product (Macedo and Silva, 2014).

The user experience unfolds as a sequence of activities: starting with the goal of interaction, followed by the formation of intentions, and then moving into the planning and execution of actions (Stanton, 1997). However, errors can arise at various stages of this sequence. For instance, users may misperceive, misinterpret, or misjudge the appliance's current state, leading to a flawed understanding of its functionality. Additionally, users might forget a specific action that should be incorporated into the sequence, or they could execute an unintended action (Stanton, 1997). Addressing these errors requires careful design choices, such as clear feedback mechanisms and intuitive layouts, to guide the user through the interaction process effectively.

The importance of user experience design in home appliances cannot be overstated. In fact, it plays a crucial role in determining both the market success of a product and the satisfaction of its users. Park et al. (2018) developed a comprehensive web-based system for evaluating UX, which helps appliance manufacturers assess and refine their products systematically. Their study highlights three key aspects of UX design for home appliances: usability, affect, and user value (Park et al., 2018).

Usability is central to ensuring an appliance is intuitive, efficient, and satisfying to use. This includes providing clear and accurate information about the appliance's status or completed actions, minimizing the effort required for users to learn how to operate it, even for first-time users, and streamlining tasks to reduce cognitive and physical effort. Ergonomic design also plays a critical role, as the appliance should be physically suitable and comfortable for users (Park et al., 2018).

Affect, on the other hand, focuses on the emotional response the appliance's design evokes. Positive emotional experiences greatly impact user satisfaction. Aspects like attractiveness, comfort, harmoniousness, and a sense of luxury or delicacy in the design contribute to creating a favorable emotional connection. Appliances should appeal to users' aesthetic preferences, provide a sense of ease, and feel coherent with the surrounding environment, whether other appliances or the overall home interior. Furthermore, high-quality materials and finishes can elevate the product's perceived value (Park et al., 2018).

Finally, user value encompasses the practical and long-term benefits provided by the appliance. This includes safety, hygiene, durability, performance, and overall functionality. Home appliances should be designed to minimize safety risks, ensure easy cleaning and maintenance, and perform reliably over time. A product that excels in these areas will meet or exceed user expectations and contribute to long-term user satisfaction (Park et al., 2018).

By integrating these principles into the design process, manufacturers can ensure that appliances not only fulfill functional needs but also meet users' emotional and cognitive desires. This approach ensures that the product offers both immediate and lasting value.

Moreover, home appliance design must take into account the diverse needs of its users. Appliances are often shared by multiple people within a household, who may differ in age, physical abilities, and familiarity with technology. As such, the design must be intuitive and flexible to accommodate these varying user contexts (Macedo and Silva, 2014). This calls for inclusive design strategies that prioritize accessibility and ease of use for all users, regardless of their physical or cognitive capabilities.

Another critical consideration is the long lifespan of home appliances. These products are used repeatedly over extended periods, meaning users may encounter a wide range of situations and challenges throughout their lifecycle. Designers must, therefore, balance short-term usability with long-term durability and adaptability to evolving user needs. A product that maintains reliable performance over time can significantly enhance user satisfaction and brand loyalty (Macedo and Silva, 2014).

In conclusion, user experience design plays a pivotal role in the development of home appliances, influencing both functional performance and emotional satisfaction. By focusing on usability, emotional appeal, and user value, designers can create appliances that not only meet immediate needs but also foster long-term relationships with users. This comprehensive approach to design ensures that home appliances remain relevant, accessible, and satisfying throughout their lifecycle.

Chapter 3

Branding

Aarke is a brand with a well-defined identity and strong values. To ensure that the continued work aligned with their vision, a comprehensive brand analysis was conducted.

3.1 Brand Identity

As previously mentioned, Aarke is a Swedish-founded company that has been in operation since 2013. The founders, Carl Ljungh and Jonas Groth, both industrial designers by trade, identified a gap in the market for carbonators as there were no machines that seamlessly combined both functionality and aesthetics. Determined to change this, they introduced the Carbonator 1.

From the very beginning, they set a clear principle: they would only expand into new product categories if they believed they could offer something superior to what was already available (aarke, n.d.-b). Today, their product range has grown to include a purifier, kettle, coffee maker and grinder, glassware and bottles, CO_2 cylinders, and the Carbonator Pro. In addition to this, they also offer spare parts and, through a collaboration with Cervera, provide a convenient CO_2 cylinder exchange service in Cervera's stores.

Aarke describes their vision as creating meaningful moments in everyday routines, contributing to a sustainable lifestyle, and redefining the ordinary. They work towards this vision by designing products that are not only easy and enjoyable to use but also aesthetically pleasing (aarke, n.d.-c). To describe what they do in a single concept, one could say engineering artistry.

The company's values are deeply embedded in their merchandise, reflecting a strong commitment to functionality, aesthetics, quality, sustainability, and innovation. It is evident from their products that everything they create is deliberate and meticulous, featuring sleek and timeless design with a touch of inspiration from Italian cuisine.

To finalize the analysis of Aarkes brand identity, an moodboard portraying what they represents in context of this project was made and is visualized in Figure 3.1.



Figure 3.1: Aarke Moodboard in context of this project

3.2 Target Group

Aarke describes their target group as "Conscious Progressive", meaning people who are meticulous about their purchases and who do not shy away from paying a higher price when they find what they want. They value both the design and function of the product and seek to live a sustainable life by choosing high-quality, sustainable materials and avoiding single-use plastic filters, water bottles, aluminum capsules, and other disposable items. People who identify with Aarke appreciate distinct yet sophisticated objects that capture attention while still fitting seamlessly into a minimalist home. For them, using Aarke's products brings both meaning and beauty to everyday tasks (aarke, n.d.-c). Three picture representing part of Aarkes target group can be seen in Figure 3.2.



Figure 3.2: Aarke Target Group

3.3 Design Guidelines

Design guidelines based on Aarke's current product portfolio were developed to ensure that the new product aligns with the company's existing design language and interaction principles. Additionally, these guidelines help maintain a consistent user experience and strengthen the brand identity.

After investigating Aarke's products, their design philosophy can be decrbibed as *Timeless*, *Minimalistic*, and *Refined*.

The colours that Aarke uses on their products are mainly steel, followed by black, sand, copper, grey and white. The metal surfaces feature polished finishes, while the colored variants have a structured, matte texture. All Aarke products are crafted from stainless steel, combined with materials such as silicone, glass, BPA-free plastic, and ocean-bound plastic. This careful selection of materials not only enhances durability and aesthetics but also aligns with the brand's commitment to quality and sustainability.

By analyzing existing products from Aarke, it has been possible to identify and understand key elements of the company's design language. Aarke's design language is characterized by clean lines, geometric precision, and a balanced combination of soft curves and sharp edges. The products exhibit a minimalist and streamlined aesthetic, where every detail serves a functional purpose while maintaining an elegant and timeless appearance. The use of high-quality materials,

along with carefully considered proportions, contributes to a sense of refinement and durability. Their design language is illustrated in Figure 3.3.

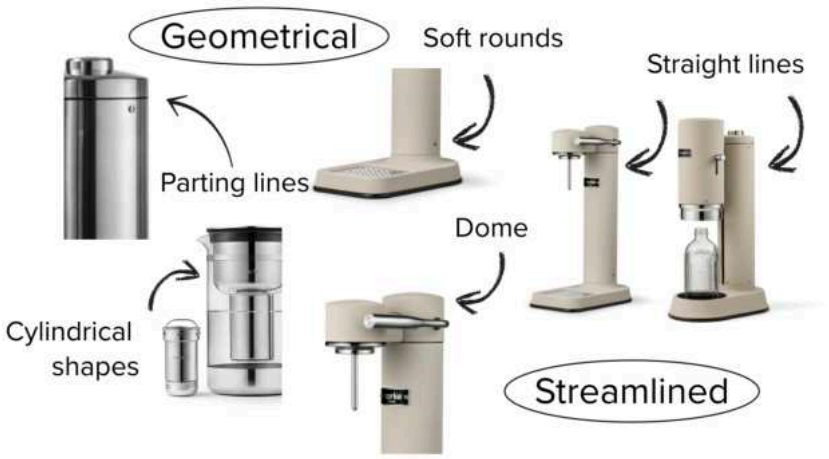


Figure 3.3: Aarke Design Language

An analysis of the interaction design across the Aarke product range reveals that different products implement varying approaches. For example, the methods used to power on devices differ, with various types of buttons and mechanisms employed. These variations illustrate Aarke's flexible approach to interaction design while still maintaining consistency within the overall visual design language. However, from a usability perspective, this inconsistency in interaction patterns challenges Norman's principle of consistency, potentially requiring users to relearn interactions across different products. Examples of these interaction methods are illustrated in Figure 3.4.

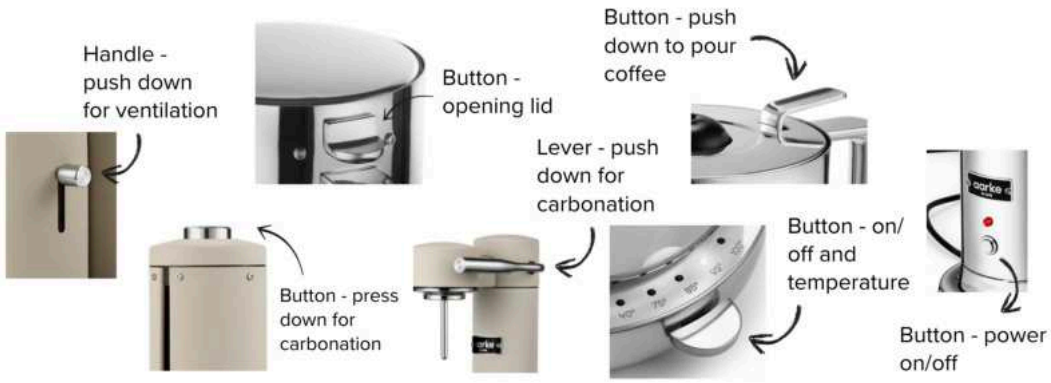


Figure 3.4: Aarke existing Interaction Design

In terms of graphics and symbols, Aarke adopts a subtle and minimalistic approach. Their products typically avoid the use of icons or symbols, relying instead on simple graphical elements such as numbers, thin lines, and similar understated details. Examples of this graphical style are shown in Figure 3.5.

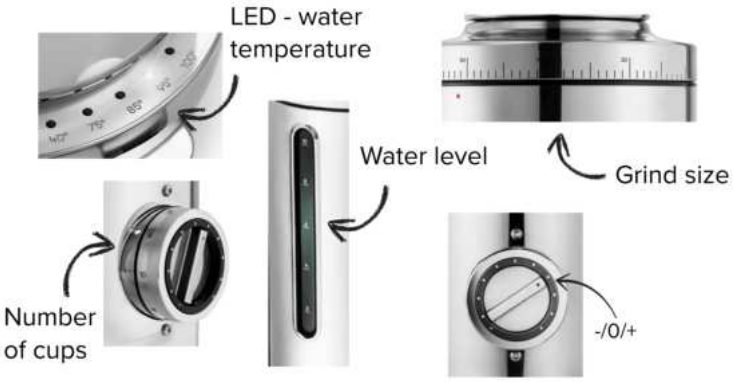


Figure 3.5: Aarke Graphics

These design guidelines will serve as a foundation throughout the project to ensure that the new product aligns with Aarke's existing portfolio. Emphasis will be placed on maintaining consistency with Aarke's established design principles, ensuring that the product seamlessly integrates into the brand's aesthetic and functional language. Using design methods already in use at Aarke, the goal is to create a cohesive product portfolio that reflects the core values and vision of the brand.

Chapter 4

Market Analysis

For Aarke products to continue to excel in both function and aesthetics, it is crucial to understand the market landscape, how it evolves, and the key selling points of competitors. Therefore, an analysis of the market for carbonators, its key players and future development was undertaken. To gain a comprehensive overview of the carbonator market, both glass and plastic bottle systems were analyzed.

4.1 Market Overview

The world market for carbonators has expanded significantly in recent years, driven by increasing consumer demand for home carbonation as a sustainable and cost-effective alternative to store-bought carbonated beverages (Tech Visionary Insights, 2024). The market growth is also supported by an rising awareness of the health risks associated with high sugar consumption and the use of artificial sweeteners (Grand Review Research, n.d.).

Sweden's carbonator market has also experienced significant growth over the past few decades. SodaStream has remained a market leader, having sold over one million carbonators to Swedish households by 2010 (Jonsson and Strandberg, 2010). As of 2015, they had captured 53% of the Swedish carbonated water market, as the trend of home carbonation and water flavoring grew significantly, putting the company in direct competition with soft drink manufacturers (Cision News, 2015). Sales continued to rise in the first half of 2016, with a 46% increase, reflecting a growing demand for sustainable and convenient everyday solutions (Mölne, 2016).

The global soda maker market is expected to continue its growth trajectory, with a compound annual growth rate (CAGR) of 7.5% from 2023 to 2030. The growth

is driven by increasing consumer preference for home carbonation systems, offering both sustainability and cost-effectiveness compared to purchasing bottled carbonated beverages (Grand Review Research, n.d.).

Manual soda makers dominate the market, holding a 74.18% share in 2022. These devices are preferred for their simplicity and the ability to adjust carbonation levels to individual preferences. However, electric soda makers are gaining popularity, as their automated process ensures consistent carbonation levels with ease. These models often come with additional features, further enhancing their appeal. As a result, demand for electric soda makers is expected to increase in the coming years (Grand Review Research, n.d.).

Manufacturers are also innovating by developing carbonators that can handle a wide variety of liquids, from flavored waters to fruit juices, to meet the growing diversity of beverage preferences. The beverage carbonator market is expected to see continued technological advances, an increase in sustainable and eco-friendly solutions, and a greater adoption of customized on-demand beverage options, which will shape the future of the industry (Tech Visionary Insights, 2024).

These market insights provide valuable considerations for the development of a carbonator from an interaction design perspective. The balance between manual control and automation highlights the importance of designing for both flexibility and ease of use. In addition, the growing demand for sustainable and customizable solutions underscores the need for an intuitive user experience that aligns with the principles of environmentally conscious design. Understanding these trends will help ensure that the final product meets consumer expectations while remaining competitive in an evolving market.

4.2 Competitor Analysis

The carbonator market, which is still a relatively small industry in terms of the number of companies, includes key players such as SodaStream, Aarke, Drinkmate, and Mysoda. By analyzing their approach to functionality, pricing, and design, valuable insights can be gained on how to further develop the Carbonator Pro.

SoadaStream
SodaStream, a company focused solely on carbonators, has been in the soda making business since 1903, and today their products are sold in 46 different countries. They offer seven different carbonators across five models, with the *Terra* and *Duo* models being available in both manual and electric versions. The remaining models, *Enso*, *Art*, and *Gaia*, are solely manual and cordless. Of the five models, only *Duo* uses a

glass bottle. All models are shown in Figure 4.1. Beyond carbonators, SodaStream also offers CO_2 cylinders, flavored drink mixes, and different bottles and accessories (SodaStream, n.d.).



Figure 4.1: SodaStream Carbonators (SodaStream, n.d.)

All models, except for the *Art* model, feature a button at the top of the device that is pressed to carbonate the water. The *Art* model, however, uses a lever for that same function, with the design choice intended to give a more retro-inspired aesthetic (SodaStream, n.d.).

SodaStream employs three distinct mechanisms for bottle attachment across its product lineup. The *Art*, *Enso*, and *Terra* models utilize a unique snap-and-lock technology to secure the bottle in place. In contrast, the *Gaia* model employs a screw-on attachment method, while the *Duo* model features a container-based system for securing the bottle during carbonation (SodaStream, n.d.).

Another feature that SodaStream incorporates is their patented Quick Connect system for attaching the CO_2 cylinder to the carbonator, eliminating the need for a screw-in mechanism (SodaStream, n.d.). The price range is between 799-2190 SEK, where *Gaia* is the budget model and *Enso* the most exculsive (SodaStream, n.d.).

Compared to Aarke, SodaStream uses a significant amount of plastic, often in black or white, in their designs, giving their products a more budget-friendly yet still trustworthy and capable image. The *Enso* model is the only one made entirely of stainless steel, which gives it a more premium, exclusive look.

Mysoda
Another company specializing exclusively in carbonators, is Mysoda, a Finland-based brand. Their carbonators are sold in several countries in Europe and the product lineup includes five different models, *Woody*, *Glassy*, *Ruby*, *Ruby 2*, and *Toby*, see Figure 4.2 (Mysoda, n.d.).



Figure 4.2: Mysoda Carbonators (Mysoda, n.d.).

The *Glassy* model is the only Mysoda carbonator designed for use with a glass bottle, while the others are compatible exclusively with plastic bottles. All carbonators are cordless and operates without electricity. Additionally, they are equipped with a standard screw-in CO_2 cylinder and a quick-lock bottle mechanism, allowing for fast and effortless bottle attachment with a short twisting motion (Mysoda, n.d.).

One aspect that sets Mysoda apart from other companies is that three of their products - *Woody*, *Glassy*, and *Toby* - are made from 100% renewable wood composite, making them the first in the industry to incorporate this environmentally friendly material (Mysoda, n.d.). *Ruby*, on the other hand, is an aluminum carbonator that is now available in an updated version with a redesigned surface that is scratch-resistant. Additionally, Mysoda incorporates an anti-freeze nozzle in their two *Ruby* models and in *Toby*, enabling the carbonation of cold water (Mysoda, n.d.).

The price range of Mysoda's carbonators varies from 702 SEK for the *Woody* model to 2681 SEK for the *Glassy* model (Mysoda, n.d.).

Mysoda's use of environmentally friendly materials and its round, simplistic design language conveys a sense of being caring and user-friendly. In terms of colors, they

offer a few different options, including pink, grey, and white. However, to maintain the sense of calm and care, the shades are fairly muted.

Drinkmate
Another leading brand in the home-carbonation system market is Drinkmate, produced by I-Drink Products Inc. Drinkmate orginates from Michigan, USA and offers different carbonators, exchangable CO_2 cylinders, syrups, and various accessories. The range consists of two countertop carbonators, one portable machine, and a water bottle that carbonates water (I-Drink Products Inc, n.d.). All models are shown in Figure 4.3.



Figure 4.3: Drinkmate Carbonators (I-Drink Products Inc, n.d.).

The distinguishing feature of Drinkmate, compared to the previously mentioned brands, is its capability to carbonate a wide range of beverages, rather than being limited to water. This is possible with their patented Fizz Infuser technique that can vary the pressure release speed (I-Drink Products Inc, n.d.).

The model *OmniFizz* carbonates using plastic bottle, and *Lux* uses bottle made of stainless steel which offers insulation to keep the water cold. The *Spritzer Portable* model is a portable carbonator that you can have both in your hand and as a countertop product by adding a stand.

Drinkmate stands out in the market, not for its aesthetically pleasing design, but for its innovative and forward-thinking products. For them, functionality takes priority over design.

There are also other, a bit smaller, players on the market that offer carbonators, including Bubliq, AGA, Stelton, and Sage. Bubliq offers one carbonator that has been recalled "*Worlds slimmest carbonator*" and their scandinavian design has been awarded (Bubliq, n.d.). AGA, also known as Linde, is a company focused on gas. Within their Aqvia concept AGA offers three different types of carbonators, *Exclusive*, *Fusion*, and *Balance* (AGA Linde, n.d.). When looking at Stelton and

Sage they visually have a resemblance to Aarkes carboantors, focusing on the aesthetics and working in stainless steel. These are also in the higher price range of the market. These products are shown in Figure 4.4.



(a) (b) (c) (d) (e) (f)

Figure 4.4: (a) Bubliq Carbonator (Bubliq, n.d.); (b) AGA Aqvia Exclusive (AGA Linde, n.d.); (c) AGA Aqvia Balance (AGA Linde, n.d.); (d) AGA Aqvia Fusion (AGA Linde, n.d.); (e) Stelton Brus Stelton, n.d.; (f) Sage the InFizz Fusion (Sage, n.d.)

4.3 Rechargeable Kitchen Appliances

A key trend identified in the market for electrical kitchen appliances was the rise of rechargeable, cordless products. This trend was investigated to understand its relevance to user convenience, mobility, and evolving consumer expectations.

A wide range of electrical kitchen appliances are available on the market. Brands such as KitchenAid, Makita, Severin, and Cuisinart offer a variety of rechargeable cordless products, which are becoming increasingly popular due to their convenience, portability, and performance. KitchenAid has a variety of cordless products in their KitchenAid Go series, including a blender, mixer, chopper, and coffee grinder.



(a) (b) (c) (d)

Figure 4.5: (a) KitchenAid Go Blender (KitchenAid, n.d.); (b) Makita Kettle (Makita, n.d.); (c) Severin Mixer (Severin, n.d.); (d) Cuisinart Mixer (Cuisinart, n.d.)

The market research on electrical kitchen appliances indicated that most products are traditionally powered via a cord connected to an electrical outlet. However, cordless appliances are becoming increasingly popular, reflecting a growing trend and demand for greater flexibility and convenience in the design of kitchen products.

4.4 Aarke Compared to the Market

Both of Aarke's carbonators are constructed from stainless steel, ensuring durability and a high-quality feel. Neither model is electrically powered, allowing users to place them anywhere without the constraint of power outlets. Additionally, both machines are designed for one-handed operation. In the Carbonator Pro, this is achieved through the innovative Push-to-Lock mechanism, which simplifies the carbonation process.

Aarke distinguishes itself on the market by focusing on aesthetics and exclusively offering premium carbonators. The company provides only two models, both designed with a balance of functionality, aesthetics, and safety. Unlike competitors that offer a broad range of products targeting various customer segments, Aarke caters to a specific market niche. Their carbonators are positioned in the higher price range, reinforcing their premium branding.

From a design perspective, Aarke emphasizes minimalism, creating products intended to be visually appealing and suitable for display on kitchen counters. The design is subtle, with no visible graphics or prominent visual features. Identifying a gap in the market, Aarke aimed to address the demand for a high-end carbonator with a focus on both aesthetics and performance.

After conducting market research and investigating competitor brands, a brand positioning analysis was developed to illustrate how Aarke and selected competitors can be interpreted in terms of their placement within the carbonation market.

Figure 4.6 illustrates this mapping, based on two axes: "design" versus "functionality" on the horizontal axis, and "premium" versus "budget" on the vertical axis. The "design" axis refers to the degree to which visual aesthetics, form, and material quality are prioritized, while "functionality" relates to the breadth and depth of technical features and performance. The "premium" versus "budget" axis reflects the product's price point in relation to perceived quality and brand positioning.

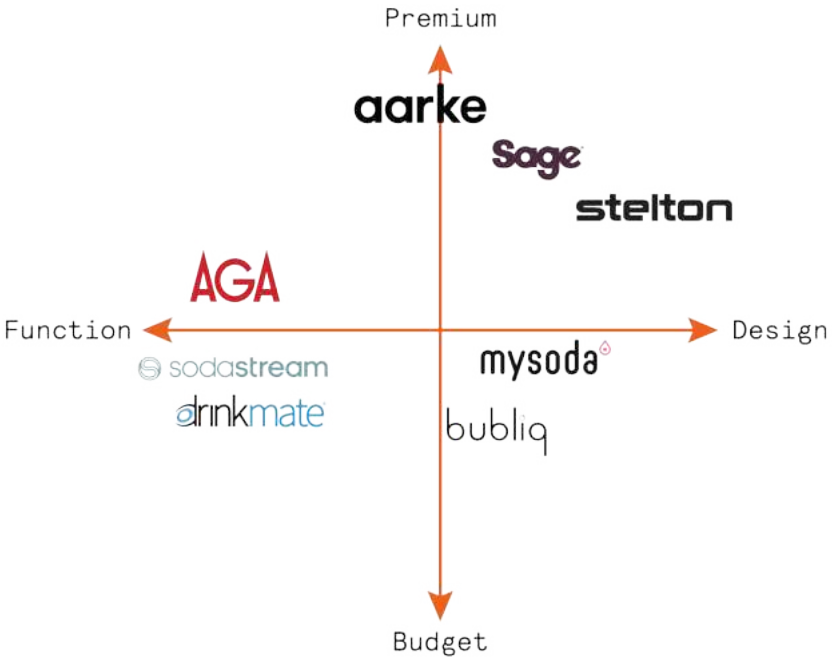


Figure 4.6: Positioning Map

According to this analysis, Aarke is positioned in the upper-right quadrant, indicating a focus on both premium quality and strong design values. Other brands such as Sage and Stelton are placed nearby, suggesting a growing segment in this space. When

Aarke first entered the market, however, it stood alone in this position. The presence of new players with similar values highlights the influence Aarke has had on defining the landscape of design-oriented carbonation products.

Aarke continues to prioritize innovation, exploring new technologies and design improvements to enhance functionality and user experience while still remaining focused on a sleek design.

Chapter 5

Product and Interaction Study

Reading about the different carbonators provided a useful overview, but to truly understand them, they had to be tested and used. For this reason, three separate studies were conducted. The first study involved the disassembling and reassembling of the Carbonator Pro. This process aimed to gain insights into its design, the technology used, how Aarke approaches materials and optimization, what features could be preserved in the new iteration, and what aspects could be improved.

The Carbonator Pro was then compared to six other carbonators, including models that use both glass and plastic bottles. The purpose of this comparison was to assess how the different products felt during use, identify the interaction points, and analyze the various solutions employed by the different models.

Lastly, a broad yet non-exhaustive analysis was conducted on products that exemplify both good and poor interaction design in general.

5.1 Carbonator Pro Study

This study involved analyzing the current design, including functionality, and user interaction. By studying this specific model, valuable insights were gathered regarding its construction, key features, and interaction design. The main objective was to understand how the Aarke Carbonator Pro operates and how users are intended to engage with it.

The interaction with the current Carbonator Pro model can be divided into eight different steps; preparation, placing bottle, activating bottle cover, carbonation, ventilation, inactivating the cover, followed by taking out the bottle. The last step

that occasionally occurs is to replace or insert the gas cylinder. Interaction steps 1-7 are visualized in Figure 5.1.

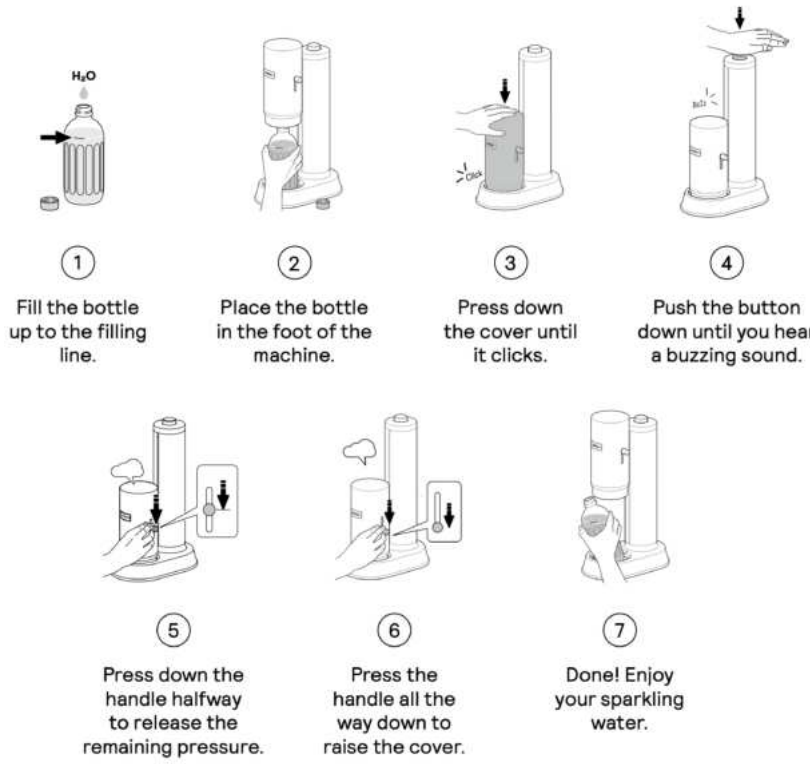


Figure 5.1: Current interaction of Aarke Carbonator Pro

A reverse engineering analysis of the current carbonator was performed to understand the rationale behind its design, the interactions between its components, and the underlying mechanics. This process provided valuable information on the standard components used by Aarke, their approach to material selection, and the degreee to which Design for Disassembly (DfD) principles were considered inproductlopment.

One key finding was that the overall size and design of the carbonator is heavily influenced by the internal mechanisms that need to be accommodated. Each design element is associated with a corresponding mechanical system that occupies space within the machine. For instance, it was observed that the height of the bottom section is dictated by the locking mechanism responsible for securing the cover.

Furthermore, the analysis revealed that the Carbonator Pro is not designed for easy repair or disassembly. DfD principles do not appear to have been a priority in

its design. Some components required specialized tools for separation, and once disassembled, it was not immediately evident how the individual parts should be reassembled. This suggests that the product is not intended to be repaired when a component fails.

The reverse engineering process also provided a deeper understanding of the carbonation mechanism and the interactions occurring within the device. From the frame of reference, interaction points should be placed close to where they provide feedback for a more intuitive design (Lidwell et al., 2003). However, this reverse engineering has shown that this wasn't always possible. That's why the locking handle ended up on the cover, making it look like a button you should press to lower it. It was also noticed that the cover only unlocks when the handle is fully pushed down.

The analysis revealed that the carbonator's outer shell is constructed from stainless steel, while the internal components are primarily made of plastic. The design approach focuses on minimizing material thickness and reducing excess material wherever possible. The only standard components identified within the device were screws, springs, and wires connected to the gas system.

Overall, the reverse engineering process provided valuable insights into the design choices made for the Carbonator Pro. While efforts have been made to optimize material usage, the lack of consideration for Design for Disassembly limits the product's reparability and sustainability. These findings highlight potential areas for improvement in future iterations, particularly in enhancing modularity and ease of maintenance.

Interview Customer Support

An interview was conducted with Monika Skogbäck, Customer Experience Specialist at Aarke, who oversees customer support for the carbonators. The discussion provided valuable insights into common customer inquiries and issues related to the carbonators, shedding light on typical challenges during setup, operation, and maintenance. These insights were essential for understanding user experiences and identifying potential areas for improving the product's usability.

The most frequent issue reported to customer support involves problems with the locking mechanism of the carbonator. Specifically, customers experience difficulty with the locking ring, where the cover does not properly lock when pressed down. Another recurring issue is the cover not lifting fully when opened. Both of these

issues were traced back to manufacturing errors, which have since been addressed and resolved.

Additionally, customers often inquire about the amount of water that collects in the foot of the carbonator. This issue is generally caused by overfilling the bottle, pressing the handle too quickly during ventilation, or a misplaced gasket. Another cosmetic issue reported by customers is the tilting of the carbonation button, which can sometimes result in the button not responding due to its misalignment.

Common customer questions also include inquiries about how to clean the carbonator, whether it contains plastic components, the differences between the two models offered by Aarke, and proper storage recommendations.

Several design-related concerns were mentioned by customers, particularly regarding the size of the carbonator. Many users perceive it as too large, particularly due to its height, which makes it difficult to fit on counters. Another notable design flaw is the difficulty in repairing or disassembling parts of the carbonator, such as the gasket, which cannot be replaced.

On the positive side, customers have expressed high satisfaction with the overall user experience. There have been no major complaints regarding the interaction design. Many users appreciate the ease of use, especially the ability to operate the carbonator with one hand. The dishwasher-safe bottle, the carbonator's robust build, and its aesthetic appeal have also been well-received.

However, some user errors have been noted. The most common mistakes involve pressing the handle too quickly during ventilation or attempting to close the cover using the side handle. Skogbäck observed that many customers do not read the manual, and regardless of how intuitive the design may be, users often find ways to make mistakes. Despite these challenges, once users receive an explanation, they typically become positive and confident in using the product.

5.2 Interaction Design in Existing Carbonators

A comparative study was conducted based on hands-on testing of six different machines. The purpose of the study was to identify all interaction points when using different carbonator models, evaluate their effectiveness, and analyze the various types of feedback employed. The evaluation was guided by insights from our literature study, ensuring that the findings were assessed in relation to established design principles. The goal was to identify key design principles, assess their impact on user experience, and determine how they should be applied in the new iteration.

The assessment focused on key aspects such as bottle attachment, bottle protection, the carbonation process, and the replacement of the gas cylinder.

Each machine was tested twice, individually by each project member, to ensure consistent evaluation and to gain two separate impressions of the interaction. Interaction points were then assessed based on Norman (1988) and his six principles; Visibility, Feedback, Constraints, Mapping, Consistency and Affordances.

The selection of the models was based on their market presence, variation in design, and use of different carbonation mechanisms. The selected carbonators included four models designed for glass bottles and two for plastic bottles. The analyzed models were *Aarke Carbonator Pro*, *SodaStream Duo*, *Mysoda Glassy*, *Zoomyo Glasslife*, *Aarke Carbonator 3* and *Stelton Brus*.

In Figure 5.2, a table is presented that outlines all the interaction points along with the corresponding functions provided by each machine.

Model	Fastening bottle	Protecting bottle	Carbonation	Ventilation	Finishing	Replace cylinder
Aarke Carbonator 3	Screwing	x	Pressing down a lever	Releasing lever	Unscrewing bottle	From beneath
Stelton Brus	Screwing	x	Turning a knob to the left	Turning knob to the right	Unscrewing bottle	From beneath
Aarke Carbonator Pro	Placing	Pushing down a dome	Pushing a button	Pushing down a handle	Lifting out bottle	From beneath
Mysoda Glassy	Screwing	Closing a door	Pushing a button	Pushing a button	Unscrewing bottle	From behind
SodaStream Duo	Placing	Tilting back container, pushing down a dome, turning a latch	Pushing a button	Turning a latch	Lifting out bottle	From behind
Zoomyo Glasslife	Screwing and sliding	Closing a door	Pushing a button	Open valve and pushing a button	Unscrewing	From above

Figure 5.2: All interaction points along with corresponding function

The two machines designed for use with plastic bottles had very similar usage processes. The bottle was filled with water, secured in place by a screwing mechanism, and then carbonated. The Carbonator 3 uses a lever that must be pressed down, while the Stelton model features a knob that is turned to the left. To release pressure, the Carbonator 3 simply requires letting go of the lever, whereas with the

Stelton, the knob must be turned to the right. The gas cylinder is changed from beneath on both models.

Both machines were perceived as easy to use due to their straightforward and minimal number of steps. However, the Carbonator 3 was more intuitive at first glance, as the lever was clearly visible and already had a well-defined affordance. Additionally, the Carbonator 3 incorporated constraints by integrates both carbonation and pressure release into a single interaction point, unlike the Stelton, which requires two separate actions. This streamlined approach contributed to a more enjoyable user experience.

In terms of feedback, the Carbonator 3 also performed better. When using the Stelton model, there was no clear indication of reaching maximum carbonation, leaving an unsatisfying feeling, as it seemed like the water might not have been fully carbonated to its potential.

When it came to the machines designed for use with glass bottles, the interactions differed more significantly between models. With Aarke Carbonator Pro, the glass bottle is placed into the designated bowl. Next, the carbonation chamber is pressed down over the bottle until it clicks into place. Carbonation is initiated by pressing a button located on the rear column, with an audible sound indicating when the process is complete. To release pressure and lift the chamber, the user pulls down a lever on the right side. The gas cylinder is changed from beneath the machine.

When using the SodaStream Duo, the bottle is placed in the angled stainless steel container. The holder is then pushed inward and kept upright while the upper part is pressed down over it. To secure the bottle, a latch is pulled to the left. Carbonation is initiated by pressing a button located on the rear column. Once the process is complete, the latch is pulled to the right to release the holder, causing the upper part to lift and the bottle holder to tilt forward. Finally, the user can remove the carbonated water bottle from the holder. SodaStream Duo utilizes the Quick Connect system for changing the CO_2 cylinder, the cylinder can then be replaced from the back of the machine.

For the Mysoda Glassy model, the interaction starts with the user pushing and screwing the bottle into the threaded mount. The door is then closed by pulling it shut to enclosure the bottle. Carbonation is initiated by pressing the front of the button on the top of the machine until a hissing sound is heard. To release the pressure, the user presses the back of the button. Once the process is complete, the door opens automatically. Finally, the user unscrews and removes the carbonated water bottle. This model has a door on the back that you open to unscrew the CO_2 cylinder and replace it.

The Zoomyo Glasslife system incorporates a Fizz Infuser for carbonation. The process begins by opening the door, filling a bottle with water, and securing it onto the Fizz Infuser. The assembly is then slid onto the carbonator, the door is closed, and carbonation is initiated by pressing the top button. Once carbonation is complete, the door is opened, and the Fizz Infuser is removed. Additional carbonation can be achieved by shaking the bottle. To release pressure, the small valve on the Fizz Infuser is opened before pressing a button for full depressurization. The Fizz Infuser is then unscrewed, completing the process. CO_2 cylinder replacement involves twisting off the top cover, lifting out the cylinder, and unscrewing it. A new cylinder is attached, and the top cover is re-secured onto the carbonator.

When comparing the first step, inserting the bottle into the machine, the Carbonator Pro, and Duo clearly outperformed the other models by the use of constraints. The simplicity of merely placing the bottle into position, rather than having to screw it in, made the process much smoother and eliminated the hassle of accidentally misalignment the bottle.

Since these machines use glass bottles, a protective mechanism is required, which all four models incorporated. However, while the Carbonator Pro, Duo, and Glassy felt secure and trustworthy, the Glasslife protection system seemed flimsy and weak. This lack of sturdiness made the machine feel less reliable to use, as the protection was neither airtight after closing nor firmly attached to the machine itself.

For carbonation, all models employed the same basic mechanism, all fairly straightforward in terms of viability and affordances. However, when it came to pressure release, the approaches varied. The only model with a relatively simple release mechanism was Glassy, where a secondary button was pressed down, creating a consistency within the product. The other models required some trial and error before the user could successfully release the pressure. This was due to the features being hidden or difficult to understand. A common issue across all models was the force required to press the buttons, making interactions feel somewhat demanding.

When it comes to replacing the CO_2 cylinder, convenience plays a key role. The ability to swap the cylinder without having to turn the machine upside down or remove it from underneath makes the process significantly smoother. Both the Duo and Glassy excelled in this regard, as their design allowed for an easy and visible replacement process. In contrast, the Carbonator Pro required more effort, as their cylinders were positioned in a way that made access less intuitive. Being able to

clearly see how and where to replace the cylinder not only made the task more efficient but also ensured that users felt confident while doing so.

Overall, carbonators designed for glass bottles were more complex to use than those for plastic bottles, primarily due to the additional step of securing the bottle for protection.

The key design insights are listed below:

- **Bottle insertion should be simple and secure:** Placing the bottle in position is more convenient than screwing it in, as long as it remains stable and secure.
- **Perceived and actual safety must align:** Carbonators using glass bottles must convey a strong sense of safety through robust and visually trustworthy protection mechanisms. A flimsy or loose enclosure can undermine user confidence, even if technically safe.
- **Feedback must be immediate and unambiguous:** Buttons, levers, and other controls should provide clear sensory feedback (audible, tactile, or visual) to confirm successful operation, particularly at critical points such as starting carbonation or completing pressure release.
- **Controls should offer calibrated resistance:** Levers and buttons must balance resistance to avoid feeling overly stiff or unresponsive. Excessive force can interrupt flow, while too little can make the product feel fragile or low-quality.
- **Intuitiveness relies on visual affordances:** The physical design should clearly suggest how each part is to be used, guiding users through the correct sequence of steps without the need for instructions.
- **Strategic placement of interaction points:** Features and interaction points should stand out slightly to signal that they are meant to be used.
- **Reduce complexity through integrated actions:** Streamlining operations by combining multiple functions into a single interaction point (e.g., carbonation and pressure release) improves ease of use and reduces cognitive load.

5.3 General Interaction Design

A brainstorming session on good and poor interaction design was conducted, where personal experiences were used to evaluate the quality of various products.

Common themes identified for poor interaction designs included:

- Lack of visual indication on the product about where and how to interact.
- Difficulty understanding how to use the interaction points.
- Too few or too many functions, leading to user confusion.
- Insufficient feedback on actions taken.
- Multiple interaction points required for actions that could be performed with a single point of interaction.
- Inconsistent interaction methods within products of the same category.
- Interaction methods that conflict with previously established functions or user expectations.

Examples of products with poor interaction design discussed included sink faucets with separate controls for hot and cold water. Another example was parking brakes in certain cars, where it is unclear whether to push down or pull up the button to activate the brake. The last example was doors with locks that require both turning and pressing the handle simultaneously. Users often need to adopt a specific grip or perform a two-step action to open the door, which can be confusing if the mechanism isn't clear. Examples are shown in Figure 5.3.



Figure 5.3: Examples of products with considered poor interaction design

Examples of products with poor interaction design included sink faucets with separate controls for hot and cold water, parking brakes in certain cars where it's unclear whether to push or pull the button to activate the brake, and doors with locks requiring both turning and pressing the handle simultaneously. Users often need to adopt a specific grip or perform a two-step action to open the door, which can be confusing if the mechanism isn't intuitive.

Regarding good interaction design, the following common principles were highlighted:

- The design mimics real-world expectations.
- The interaction method follows established interaction norms.
- Clear, immediate feedback is provided, confirming to the user that their action was successful.
- Consistency is maintained, with similar actions using similar methods across the product, reducing cognitive load.
- Affordance is present, where the design clearly indicates how it should be used.
- The interaction process is efficient, minimizing unnecessary steps and complexity.
- A sense of control is provided to the user, ensuring that the user feel confident while interacting with the product.

Examples of well-designed products included faucets with a single handle, trash bins that open when stepped on, and hair straighteners that use color indicators to show which areas are hot, see Figure 5.4.



Figure 5.4: Examples of products with considered good interaction design

The identified examples of poor and good interaction design closely correspond with the principles outlined in the Frame of Reference. Poor designs typically lacked clear affordances, intuitive mappings, and consistent feedback, leading to confusion and inefficient use. In contrast, well-designed products demonstrated strong adherence to these principles by providing visible system status, natural and predictable control mappings, consistency, and immediate, distinct feedback. These findings reaffirm the relevance of established interaction design frameworks and emphasize how applying these principles leads to more user-friendly and satisfying experiences across different product categories.

This exercise was conducted to expand the perspective beyond carbonators, aiming to identify if the interaction issues observed in that context are shared across other product categories. The conclusion drawn was that, despite the varied domains, the interaction challenges are often consistent. These common issues include unclear or excessive functions, poor feedback, and overly complex actions or movements.

Chapter 6

User Study

A significant part of understanding the interaction design in the current model was the user study where observations were made of users interacting with Carbonator Pro.

6.1 Study Purpose and Objectives

Observations were conducted to analyze user interactions with the carbonator in a real-world environment. These observations facilitated the systematic documentation of user behaviors, offering objective data on how individuals engage with the device. The process helped identify authentic user actions, including unconscious behaviors, challenges, and instances of confusion.

According to Nielsen, it is preferred to have a small test group when investigating usability, to work iteratively and do more tests with fewer participants (Nielsen, 2012). Therefore, a group of less than ten participants was sought for.

To ensure a structured and systematic approach, an observation guide was developed, based on key principles from the literature study. The guide aimed to capture critical aspects of user interaction, including usability, efficiency, and adherence to intended design principles.

The development of the guide was informed by Norman's (1988) principles of interaction design, focusing on visibility, feedback, constraints, mapping, and affordance. In addition, Nielsen's (2012) usability components were incorporated to assess learnability, efficiency, memorability, error prevention, and user satisfaction. These frameworks provided a foundation for evaluating the intuitiveness and

effectiveness of the carbonator in real-world use. The following key principles and usability components were considered during the evaluation:

- **Visibility, Constraints, Mapping, and Affordance:** Assessed by observing how easily users identified interaction points and understood how to operate the device.
- **Feedback:** Evaluated by allowing participants to reflect on the cues received during operation.
- **Learnability and Memorability:** Measured by comparing first-time users with experienced users to assess adaptation speed and recall of the interaction process.
- **Efficiency:** Determined by timing the process from an empty bottle to a fully carbonated one.
- **Error Occurrence:** Documented throughout the test to identify common mistakes and their causes.
- **User Satisfaction:** Assessed through direct observations and post-interaction reflections.
- **Emotional Design and Engagement:** Evaluated by capturing participants' reactions and feelings before, during, and after use.

6.2 Study Design and Methodology

Based on these objectives, the study was designed to capture and assess user interactions through structured observations.

A group of seven participants for observation was assembled with the objective of creating a diverse and representative user group, including both first-time and frequent users. The observations would then both capture insights into the ease of initial interaction, while frequent users helped determine whether their actual interactions aligned with the intended usage. The participants for observation also varied in terms of gender, age, background, and interest in technology. Table 6.1 presents all participants with specifications.

Table 6.1: Observation Participants

Participant	Age	Gender	Experience	Interest in Technology
A	27	Female	First-time user	Not interested, used other carbonator earlier
B	24	Female	First-time user	Engineering student, used other carbonator earlier
C	56	Female	First-time user	Not interested
D	58	Male	First-time user	Not interested, never used a carbonator
E	57	Female	Frequent user	Not interested
F	49	Female	Frequent user	Engineer
G	26	Male	Occasional user	Engineer, very interested

The interaction process was divided into seven steps: filling the bottle, placing it in the machine, activating protection, carbonating, ventilating, deactivating protection, and replacing the CO_2 cylinder. For each step, feedback factors were observed and analyzed, such as ease of interaction and understanding of feedback cues. The guide consisted of two sections, one observation guide for first-time users, and the other for experienced users. The full observation guide can be found in Appendix A.

First-time Users

Step 1: The user is given an empty glass bottle and is asked to use the Carbonator Pro to sparkle a bottle of water. A think-aloud protocol will be used during the observations, where participants will be asked to verbalize their thoughts, expectations, and challenges as they interact with the carbonation machine. This approach provides valuable insights into the cognitive processes behind user interactions and helps identify potential usability issues. The person is then asked to replace the CO_2 cylinder. This step is ended with some reflective and follow-up questions regarding the experienced interaction.

Step 2: The intended usage is shown and the object is then asked to use the device again. Another reflection will be obtained discussing the intended interaction.

Step 3: After the observation session, participants will complete a Likert-scale questionnaire to evaluate their experience, see Appendix A. The questionnaire will

include scaled questions, such as 'How easy was it to understand where and how to carbonate?' rated from 1 (very easy) to 5 (very difficult). This will provide quantitative data on user perceptions and usability.

Experienced Users
Step 1: The user is asked to use the carbonator as they are used to, the interaction is documented in video. This step allows the user to do without putting to much thinking behind the operation.

Step 2: The user uses the carbonator another time, where they incorporate the think-aloud protocol and they can now start reflecting about what they do. They are further asked to replace the CO_2 cylinder.

Step 3: The last step is a reflection and discussion about the user interface and the interaction.

Below, in Figure 6.1, examples are shown of how it looked when some of the participants interacted with the carbonation machine during testing.



Figure 6.1: User Study Test

6.3 Results and Key Insights

The observations was overall successful, yielding the desired insights. The participants seemed to have found the instructions and questions easy to understand and much of what was said and done aligned with the previously identified impressions and assumptions regarding the carbonation machine.

The findings suggest a significant difference in performance between first-time users and experienced users. On average, first-time users required 1 minute and 12 seconds to carbonate a bottle of water, whereas experienced users completed the same process in just 20 seconds. This significant time difference underscores the impact of user experience on the efficiency of the interaction.

Participants generally described the carbonator as large, robust, high-quality, stylish, and luxurious. However, the design of the base was often compared to a dog bowl.

First-time Users
Regarding the interaction points, the bottle had positive and negative responses. Participants found the bottle easy to grip and appreciated the clear fill level markings. However, some ambiguity arose when determining the correct placement of the bottle. Fifty percent of first-time users instinctively attempted to push the bottle upward into the cover and secure it using a twisting motion. One commonly cited frustration was the need to tilt the bottle in order to place it properly in the base. This suggests that adjusting the constraints, such as reducing the need to tilt the bottle, would address user frustrations and improve ergonomics.

For the next step, activating the cover, participants initially attempted to interact with the side handle. Some used the handle to push the cover down, while others applied pressure directly to the hood, often requiring both hands. During the carbonation process, participants found it easy to identify the interaction points. However, uncertainties emerged regarding the operation of the carbonation button. It was unclear whether the button should be held down continuously or pressed multiple times. Additionally, users were uncertain when the carbonation process was complete and how to gauge the desired level of carbonation in the water. This highlights the need for clearer feedback, such as visual or auditory cues, to enhance confidence in the process and reduce ambiguity.

Participants found it easy to identify the handle as the main interaction point for releasing pressure and opening the cover. However, first-time users did not realize that ventilation and opening could be done in one motion. Instead, they pressed the handle multiple times to fully open the cover. Some users found the handle slightly stiff but appreciated that the cover lifted on its own.

The most confusion occurred during gas cylinder replacement. All first-time users initially tried to unscrew the top of the carbonator, mistaking the dividing line below the carbonation button for the access point. Removing the cylinder from the bottom was seen as difficult, with resistance in the screwing motion making the process even harder. Laying the machine down to replace the cylinder caused concerns about damaging the table or the carbonator itself.

Once participants learned the correct usage, they found the carbonation process straightforward. However, replacing the gas cylinder remained the most challenging aspect. Additionally, the side handle created the most uncertainty for first-time users. Making the function of the side handle more intuitive could improve the overall user experience.

Experienced Users
The experienced users who participated in the test interacted with the carbonator as intended, indicating that the process is easy to learn and becomes increasingly intuitive with repeated use. This suggests that while first-time users may encounter some initial confusion, the learning curve is relatively short and manageable.

Although the experienced users did not encounter significant issues with the machine, several points of irritation were raised. The first concern was the accumulation of excess water in the bowl where the bottle is placed. The second issue involved the need to tilt the bottle when both placing and removing it. Lastly, some users expressed a preference for a more compact carbonator, as the current size takes up considerable counter space. Addressing this concern could improve the product's overall ergonomics and appeal.

Overall, the experienced users were satisfied with the carbonator. They highlighted its ease of use, simplicity of operation, and the ability to operate it with one hand. Additionally, they appreciated the speed of the carbonation process.

In summary, the issues identified in the user study clearly reflect the presence orabsence of key interaction design principles. By addressing weaknesses in feedback, constraints, and mapping, the product could become more intuitive, especially for first-time users, while preserving the elements that already support effective interaction. The insights was concluded into a customer journey map that highlights the steps with needs and opportunities at each step of the process, see Appendix B.

Chapter 7

Project Definition

The aim of this project was to investigate how interaction design can improve the user experience of carbonators by addressing common frustrations and usability issues. The process began with an in-depth exploration of current user behaviors and a careful identification of pain points associated with the existing Aarke Carbonator Pro.

Through early research and analysis, it became evident that many of the issues users face stem from a lack of intuitive interaction and feedback. Based on the identified findings, the project scope was redefined to focus on the development of an electrically rechargeable carbonator, with particular emphasis on interaction design as a key factor in improving usability and user experience.

This section presents a summary of key insights gained during the initial phases of the project, which serve as the foundation for the new concept and design direction. This definition stage is the midpoint between the two diamonds in The Double Diamond model.

7.1 Insights

Based on the user research, surveys, and initial analysis conducted in the earlier phases, several recurring challenges and needs related to the current carbonator has been identified. These insights have shaped the understanding of user behavior and expectations and ultimately guided the project toward a new scope with a more defined project definition.

Both usability, and design-related issues were identified. The findings span across practical, aesthetic, and interactional aspects of the product experience. Key insights include:

- **Mechanical and functional failures:** Users frequently report problems with mechanisms getting stuck or breaking, particularly the locking mechanism and protective cover. This leads to frustration and a lack of trust in the product's reliability.
- **Difficult maintenance and unclear instructions:** Many users struggle with tasks such as ventilating the bottle, emptying residual water, or changing the CO_2 cylinder. Instructions are often unclear, and users tend to avoid reading manuals.
- **Lack of intuitive interaction:** First-time users often experience confusion due to insufficient interaction cues. Feedback mechanisms, visual, auditory, or haptic, are absent, making it difficult to know how much carbonation has been added or when the process is complete.
- **Poor visual and product design:** Aesthetics also affect user perception. An example is the foot of the machine, which some users compared to a dog bowl, affecting the perceived appeal.
- **Excessive size and impractical form factor:** The machine is described as bulky, taking up significant space in the kitchen. Users express a desire for a more compact and space-efficient design.
- **Unclear feedback about CO_2 levels:** Many users cannot tell when the gas is depleted. This results in misinterpretations of product failure and leads to unnecessary customer support inquiries.

7.2 New Scope

Recognizing that many user challenges, such as lack of feedback, unclear processes, and difficulty understanding the machine status, stem from the limitations of a manual system, the project shifted direction toward developing an electrically rechargeable carbonator.

Electric functionality introduces new opportunities for improved interaction design. With integrated components, the carbonator can offer enhanced feedback, clearer signifiers, and more intuitive controls. These improvements aim to create a more user-friendly and responsive experience.

Market research also supports this shift. There is strong demand for smart household appliances, and consumer expectations continue to grow. This new direction positions Aarke to align with market trends and expand into a broader product space. User feedback further revealed that the current carbonator is considered too bulky. An electric model opens the door for a more compact and space-efficient design.

This pivot of making an electric carbonator represents a strategic step for Aarke, broadening its reach and improving the product experience overall.

Continuing, the insights gathered from the current carbonator model were categorized into two areas: features to retain and aspects to further develop or enhance. Key features that will carry over to the new model include the bottle attachment method, allowing users to simply place the bottle without the need to screw it into position, and the auditory feedback during carbonation, which users found clear and helpful. The signature minimalistic design of Aarke products will also be maintained and many of the design elements will be derived from Aarkes already established products. Additionally, the carbonation process should continue to be operable with one hand and remain quick, involving only a few user steps.

Improvements will be made in the areas of interaction design, user-friendliness, and aesthetics. In terms of interaction, the focus will be on refining the mechanisms for ventilation, opening and closing the cover, and replacing the gas cylinder. Enhancing user-friendliness could involve features such as indicating when the cylinder needs to be replaced or allowing the user to select the desired level of carbonation. As for aesthetics, the primary goal is to create a more compact and streamlined machine.

The primary focus remains on interaction design, with the new direction emphasizing how electric functionality can contribute to improved interaction and overall user experience.

7.3 Requirement list

Based on the insights and direction outlined, a list of requirements was defined to guide the development of the new electrically rechargeable carbonator. These requirements was aimed to address the identified user needs, improve usability, and align with Aarke's design principles.

The specification includes both functional and non-functional requirements, each marked as either essential or desirable, along with a brief explanation. Some key requirements include:

- The process shall be operable with one hand.

- The system shall prevent carbonation if the bottle is not securely covered by the safety mechanism.
- The protective cover shall withstand the pressure generated when carbonating an empty bottle.
- The battery shall be capable of performing a minimum of 50 carbonation cycles on a single charge.
- The system shall prevent carbonation if the bottle is detected to be empty or insufficiently filled.

A complete list of requirements is provided in Appendix C.

Chapter 8

Ideation

With the project scope defined and key insights established the project enters the second diamond in the Double Diamond model. This next phase focused on generating ideas and exploring potential design directions. The purpose was to translate user needs, technical constraints, and design guidelines into early concepts for the new product.

8.1 Brainstorming

Brainstorming sessions were conducted to explore multiple directions within key areas of the concept development, specifically how the bottle could be enclosed, how the CO_2 cylinder could be replaced and interaction design aspects.

Bottle Encapsulation

It became evident early in the development phase that many design decisions, particularly those related to interaction design, were closely influenced by the initial structural layout of the carbonator. Furthermore, it was observed that the method used to encapsulate the bottle played a critical role in shaping the overall form factor of the machine.

As a result, the beginning stage of the ideation process focused specifically on exploring different solutions for encapsulating the bottle. Five distinct categories of encapsulation mechanisms were identified, and iterative sketching was conducted within each category to explore their potential.

The first category was the hood solution, inspired by the protective hood mechanism seen in the Aarke Carbonator Pro. This concept involves a retractable hood that descends to enclose the bottle during the carbonation process. Preliminary sketches of this concept are presented in Figure 8.1.

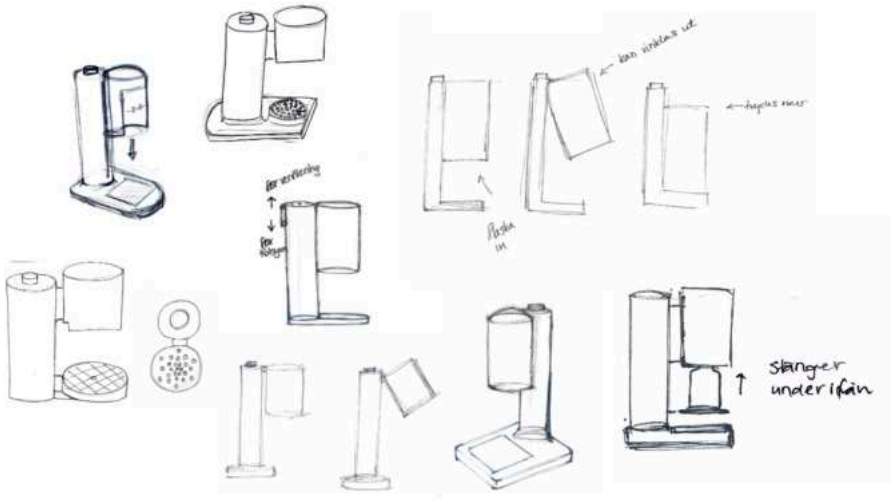


Figure 8.1: Early sketches of the hood concept

This solution category was eliminated early in the process, as it did not address the issues identified with the current Carbonator Pro. The primary drawback of this approach is that it does not reduce the height of the machine, which remains a key design concern.

The second category was the door mechanism. In this concept, the bottle is placed inside the device and then enclosed by closing a hinged door. Early visualizations of this idea are shown in Figure 8.2.

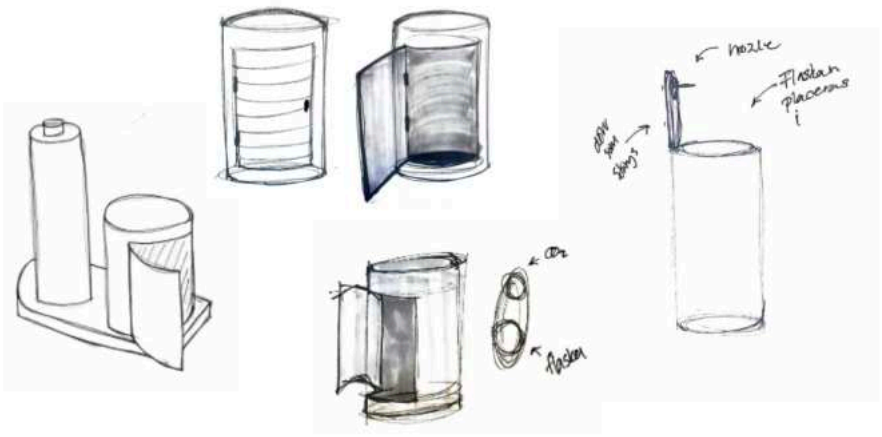


Figure 8.2: Early sketches of the door concept

The third category was the garage solution, drawing inspiration from the functionality of a garage door. This approach involves a sliding or folding panel that covers the bottle area horizontally or vertically, creating a compact and seamless enclosure, see Figure 8.3.



Figure 8.3: Early sketches of the garage concept

To explore a slimmer and more visually lightweight design, an alternative concept was developed in which the bottle is enclosed by two separate halves rather than a single door. These two parts come together to form a complete enclosure around the

bottle, contributing to a more open and airy appearance. Multiple design variations of this approach were sketched and are presented in Figure 8.4.

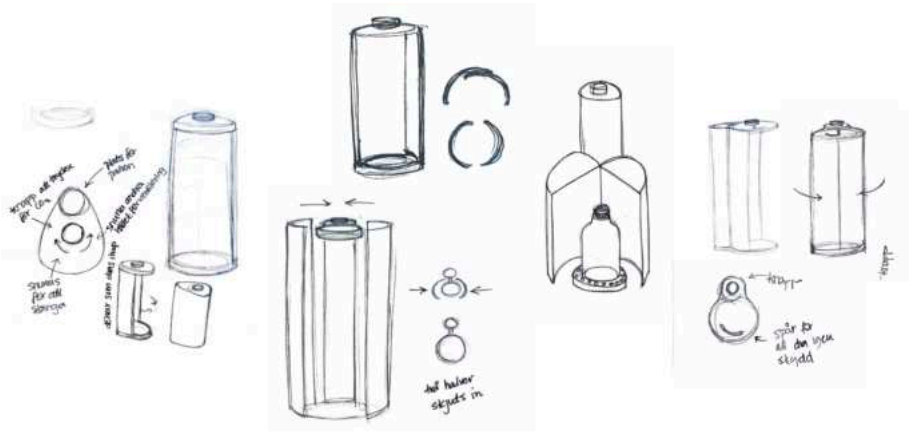


Figure 8.4: Early sketches of the two halves concept

The final category in this phase was the container solution. In this concept, the bottle is placed inside a container that functions as a sleeve, effectively encapsulating the bottle. Different approaches were explored for how to place the bottle into the container, with sketches of the different solutions presented in Figure 8.5.

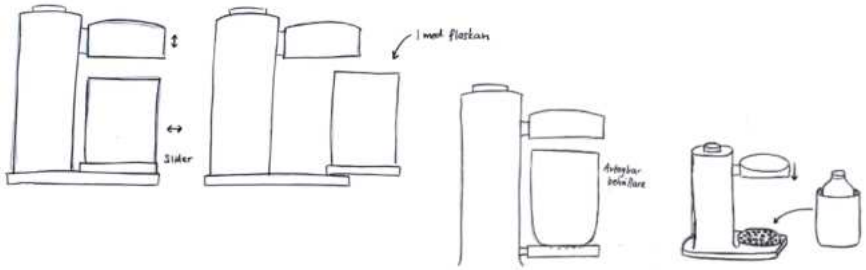


Figure 8.5: Early sketches of the container concept

The remaining categories were evaluated and compared through early prototyping. Cardboard prototypes were created for at least one concept within each category. The purpose of the prototyping was to gain a better understanding of how these ideas would appear and function in a real-world context. Prototypes are shown in Figure 8.6.



Figure 8.6: Cardboard prototypes of concepts

It was observed that all garage-style solutions involving a folding panel door being pushed downward were suboptimal, as they required substantial space in the open position and presented challenges in achieving a fully sealed enclosure. As a result, this concept was not pursued further. However, the garage concept with a side-opening door remained a viable alternative. The hinged door solution, while highly user-friendly and intuitive to operate, was also eliminated because it did not convey the desired sense of safety. Additionally, a hinged door has more closed surfaces when the carbonator is not in use, leading to a bulkier and more heavy feel. Regarding the two halves category, the double sliding door was chosen for further evaluation since it created a slim design. Lastly, the container solution was also perceived as a viable solution that could reduce the total height of the machine.

In the end three concepts of bottle encapsulation were chosen; the single sliding door, the double sliding door, and the container. These three concepts were further evaluated and compared. Rough CAD models were put together to further develop insight into how these concepts would function.

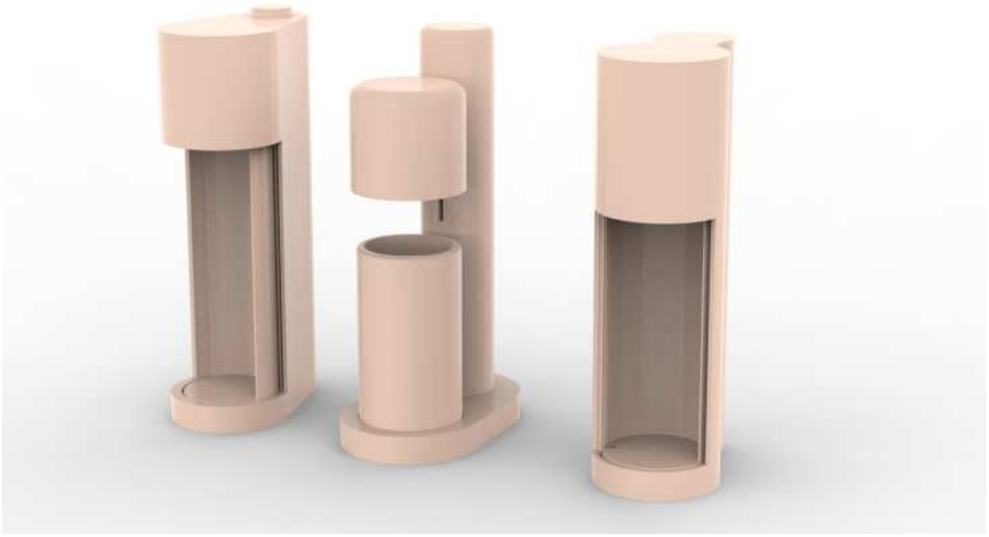


Figure 8.7: Rough CAD models of the three bottle encapsulation concepts; the double sliding door, the container, and the single sliding door.

Cylinder Replacement

Another part of the carbonator that also was ideated on in this early step was the method of replacing the CO_2 cylinder. The focus was on simplifying this process, which involved avoiding replacing the cylinder from underneath. The methods that remained were then to replace the cylinder either from behind or from above. Sketching was done on various solutions, see Figure 8.8.



Figure 8.8: Early sketches of replacement of CO_2 cylinder.

For solutions involving CO_2 cylinder replacement from behind, the goal was to design a discreet yet intuitive opening and clearly indicate how to access the back. Inspiration was drawn from Aarke's existing product line and established interaction points. Various methods to assist with unscrewing the cylinder after opening were also explored. Concepts for top-loading replacement were likewise developed, focusing on clear user guidance for integration and access. However, a challenge with this approach is the potential conflict with components placed above the cylinder in the rear column, which can complicate the design.

These solutions could be integrated into any of the bottle encapsulation concepts.

Interaction Design

Even though the interaction design was closely tied to the chosen method of bottle encapsulation, early ideation was conducted to explore possible interaction solutions and assess their feasibility within the different concepts. The focus was on designing intuitive and meaningful user interactions for the critical steps of the carbonation process, primarily initiating carbonation and ventilating. While the bottle encapsulation concepts addressed the opening and closing of the protective cover, additional attention was given to creating clear, engaging, and physically satisfying interactions for the remaining user actions. Considerations also included semiotic elements and feedback mechanisms to enhance the overall user experience and communicate system status effectively.

A key ambition in the development of the carbonation and ventilation interaction was to create a distinct mechanical experience. The goal was to avoid simple touch-based inputs in favor of an interaction that the user can physically feel and connect with, both through motion and auditory feedback. A mechanical action helps convey that something is actively happening, such as the carbonation being injected, and provides a sense of control and clarity. Based on this intent, various mechanical solutions were explored, including levers, buttons, and rotary knobs. Inspiration was drawn from existing Aarke products, which feature both a lever and button mechanism, as well as from other familiar interaction types commonly found in kitchen appliances. These options were evaluated for their ability to reinforce the user's perception of action, function, and feedback in a clear and intuitive way.

In parallel with the exploration of mechanical interaction methods, feedback and semiotics were also key considerations. Various visual indicators were ideated, such as LEDs, progress bars, and symbols, to signal states like active carbonation, process completion, or the need to replace the CO_2 cylinder. Particular attention was given to the placement of these indicators, following the principle of mapping. Indicators were placed near the components or actions they referred to, ensuring clear and intuitive communication, for example, placing indicators for carbonation level close to the carbonation interaction point. Early sketches of interaction design is presented in Figure 8.9.

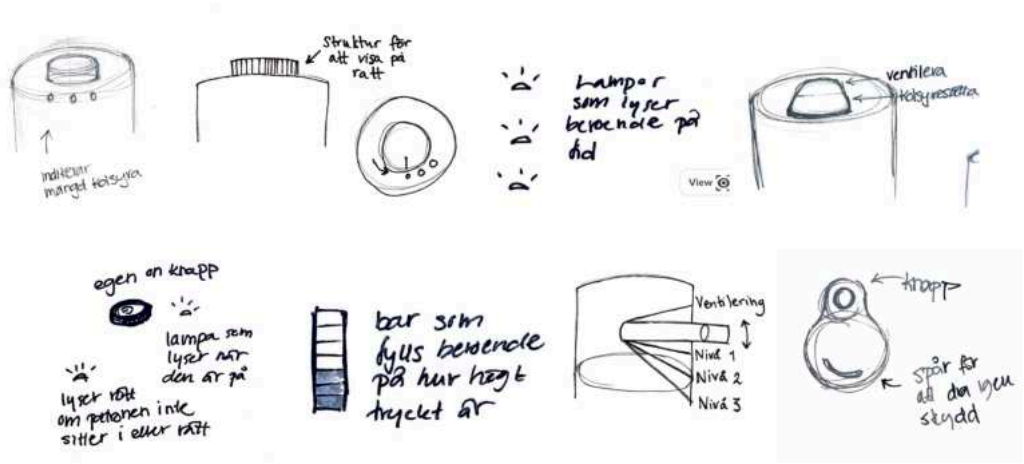


Figure 8.9: Early sketches of interaction design.

For the interaction design the mission was to create a balance between user-friendliness and intuitiveness, but still remaining minimalistic and discreet, in line with the design guidelines, see Chapter 3.

The interaction design was developed separately from the main concepts to keep it flexible and adaptable. This way, the interaction methods can work with different mechanical solutions while ensuring a smooth and intuitive user experience that fits the overall design.

8.2 Concept Selection

The initial method used to compare the various concepts and identify a preferred solution was through the development of flow charts, see Appendix D. A flow chart was created for each concept to systematically analyze the entire carbonation process, detailing user interaction and system behavior at each stage. All concepts were evaluated based on five primary steps: bottle placement, closing and sealing the cover, initiating carbonation, ventilating, and opening the cover. Each main step was further broken down into sub-steps, which varied depending on the specific concept. This approach enabled a clear comparison of the complexity, usability, and feasibility of each proposed solution.

In the first main step, the only notable difference was that the container concept included an additional sub-step. Unlike the door concepts, where the bottle is placed directly onto the carbonator, the container concept requires the user to first place the bottle into the container and then position the container onto the carbonator. In the next step, however, the door concepts require an additional user action. While the container concept combines closing the cover and lowering the nozzle into a single motion, the door concepts separate these into two steps: first closing the door(s), and then engaging a separate mechanism to lower the nozzle onto the bottle. In the subsequent steps, no significant differences were observed between the concepts in terms of user interaction. Instead, variations at this stage primarily concern the internal mechanical and electrical solutions within the product.

A weighted criteria matrix was done to further evaluate the three concepts based on criteria found during the research and user study phase. The criteria, with their weighting, and the results are presented in Figure 8.10.

	Criteria	Door	Two Halves	Container
1	Operate with one hand	3	3	1
2	Few moving mechanisms	2	1	3
3	Sleek and compact design	1	2	3
1	Easily empty spilled water	3	3	3
3	Safe	2	1	3
2	Place bottle without tilting	3	3	3
3	Follow Design Guidelines	2	1	3
2	Clear where to integrate	3	3	3
2	Clear how to integrate	3	2	3
2	Enough space for inner parts	2	1	3
2	Few user steps	2	3	1
1	Simple inner mechanisms	2	1	3
	TOTAL	53	45	66

Figure 8.10: Weighted Criteria Matrix for Bottle Capsulation Concepts.

After evaluating the concepts using both flow charts and a criteria matrix, the container concept emerged as the most effective in meeting the defined criteria and user needs. It performed well across most categories, although it scored lower in terms of one-handed operation and the number of user steps. While the process can technically be completed with one hand, it may be more practical to use both hands when placing the bottle into the container and then positioning the container onto the machine, this also accounts for the additional user step compared to the other concepts.

8.3 Chosen Concept

The chosen concept features a container where the glass bottle is placed. The container is then placed on the carbonator. The idea with the container is that it functions as a cooler when it is not used during carbonation. The container also becomes the collector of water that can be spilled during the carbonation process. The user can easily empty and clean the container. A hood, similar to the one on the Aarke Carbonator Pro, is moved downward to enclose the bottle. Then carbonation can be initiated, followed by ventilation and opening of the cover. The concept is strongly associated with the Aarke brand, with the form similar to the existing carbonators in Aarke's portfolio.

Chapter 9

Concept Development

The selected concept, featuring a container-based encapsulation of the glass bottle, was further developed in this phase through multiple iterations. The development process involved refining both the design and user interaction based on continuous evaluation.

To support this, two trigger material tests were conducted, one focusing on user interaction during carbonation, and the other addressing technical considerations. These tests provided valuable insights into user preferences and feasibility.

9.1 Design Development

Components of the carbonator was further iterated to refine the design of the product. During this phase, several design elements were drawn from existing carbonators in Aarke's product portfolio to ensure consistency with the brand's established design language. Focus was mainly on form and design language.

The Base

Design iterations for the base of the carbonator focused both on the overall shape of the foot and on how the container is placed and integrated onto it.

The design of the foot was developed through iterations from two perspectives: top-down and side views. From the top, various shapes were explored, drawing inspiration from both the Aarke Carbonator 3 and Pro models, as well as other designs that align with the two cylindrical sections above the foot. These iterations are shown in Figure 9.1, where the rear circle represents the column housing the CO_2

cylinder, and the front circle corresponds to the container. Additionally, iterations from the side view were tested to experiment with different base angles and how the underside should meet the surface on which the foot is placed.

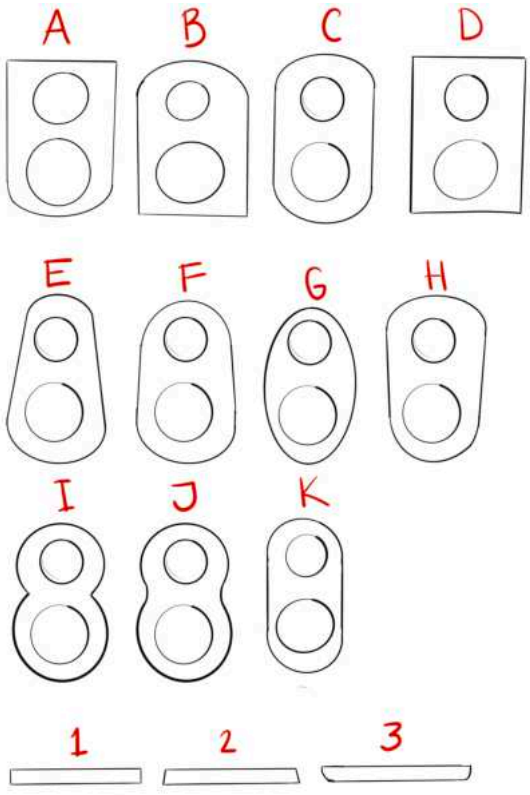


Figure 9.1: Form iterations on the base of the carbonator.

Since a rounded shape provides better visual alignment with the other components of the carbonator, and a geometric, simple form adheres to the overall design guidelines, iterations C, F, and K were selected for further development into prototypes, see Figure 9.2. Each of these form iterations was paired with a corresponding side-view concept to create complete prototype versions for evaluation, Figure 9.3.



Figure 9.2: Form iterations C, K, and F of the base.



Figure 9.3: Form iterations 1, 2, and 3 of the base.

The final form of the foot was based on iteration K, combined with the second side-view option. Iteration K offered a smooth transition between the circular and straight sections, while no clear advantages were identified with the angled approach seen in option F. The selected side-view, where the foot widens closer to the ground, was perceived to convey a sense of stability and robustness, qualities that align with the desired expression of the product.

Iterations were also made on the design for the container placement on the foot. The goal was to incorporate some form of indication to guide the user on how and where to place the container. This guidance was intended to make the process intuitive, signaling both the correct placement of the container and its ability to be removable. The proposed solutions are shown in Figure 9.4.

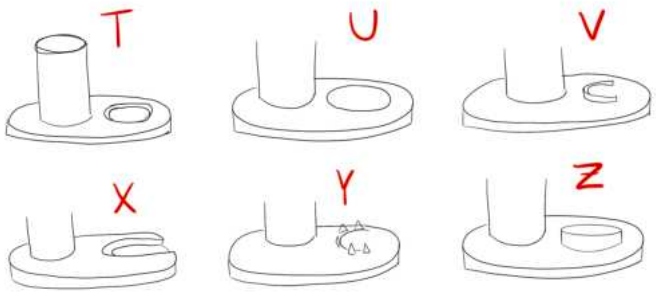


Figure 9.4: Design iteration on how to place container.

A minimalistic design approach led to the exclusion of concepts V and Y. Furthermore, option Z was disregarded due to its impact on the overall height of the product, which was to be kept compact. Concept U lacked any guiding function for the container, making it less viable. As a result, only solutions T and X were selected for further development and prototyping, as shown in Figure 9.5.



Figure 9.5: Prototypes of design iteration on how to place container.

The circular indentation in the base clearly indicated the correct placement of the container, minimizing the risk of user error. In contrast, the slot provided both a visual and functional guide for placing the container, while also suggesting its removability. Given the weight of the container, especially when filled with a glass bottle, the slot allowed for easier positioning through a sliding motion rather than requiring the user to lift and place it. As a result, the sliding slot was selected as the final solution for container placement. The radius of the slot was designed to align with the diameter of the container. The sliding motion is shown in Figure 9.6.



Figure 9.6: Sliding motion when placing container on the base.

The Protective Cover

Form iterations were carried out on the protective cover with a focus on how the cover and container should fit together when closed. These developments are illustrated in Figure 9.7.

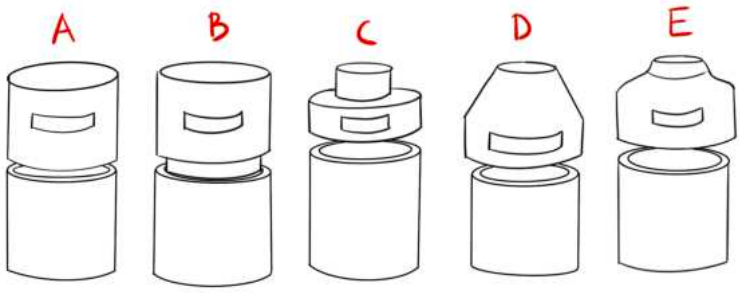


Figure 9.7: Form iterations on the protective cover.

In the end, a cylindrical shape was selected for the cover to visually align with both the container and the rear column. An inner ring of the cover, as seen in concept B, was chosen to be added for connecting the cover with the container during usage. With the inner ring ending on the inside of the container it creates a smooth transition between the two parts on the outside, resulting in the most ascetically pleasing result.

Different electrical solutions were explored for locking and releasing the cover in its lowest position. While the Aarke Carbonator Pro uses a mechanical locking ring, an electrical solution offers advantages in terms of space efficiency and a cleaner visual design, as it avoids adding visible mechanical components. After discussing with a teacher at the mechatronik chapter, a solenoid was identified as a suitable option,

intended to lock and hold the cover securely during carbonation. Once the process is complete, the solenoid would disengage, allowing the cover to automatically return to its initial position. Further technical details regarding this solution are presented in Chapter 10.

Interaction Design

The interaction point to elaborate on was: how the protective cover should be lowered, how the carbonation process should be carried out.

The same approach for lowering the cover as currently used on the Carbonator Pro, pressing down by hand, was chosen, as it was seen as the simplest and most intuitive method. It also provides a sense of familiarity for customers who have previously used the Pro model. However, a key usability issue identified in the current Carbonator Pro was first-time users not understanding the pressing action and often attempted to close the hood using the side handle. Therefore, in this new design, the cover is free from additional interaction points, which is intended to eliminate that confusion and reduce the likelihood of user error. Moreover, as the cover moves vertically along a track, the track will further serve as a clear visual cue, indicating to the user that the cover should be pushed downward.

Different types of interaction for initiating the carbonation process methods was ideated on, with inspiration from interaction methods that Aarke already utilizes in their products, see Figure 9.8. The placements shown are not the decided placements of each part.

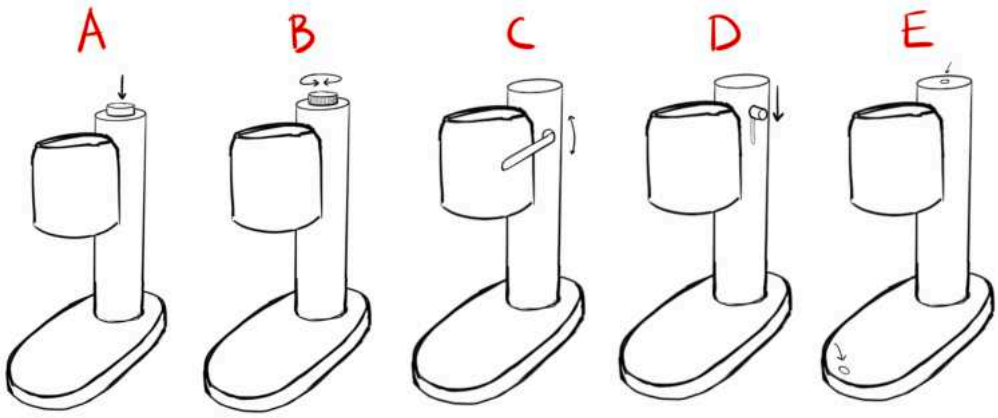


Figure 9.8: Iterations on interaction design for carbonation.

The decision to use a mechanical carbonation process had already been made prior to the start of the ideation phase, in order to ensure that the user retains direct control over the gas release during carbonation, making touch-sensitive buttons, concept E, an unsuitable option. Continuing, the side-mounted handle on the rear column (concept D), which is used in Carbonator Pro for ventilating, was not pursued further, as it risked causing confusion due to inconsistent functionality across different Aarke carbonator models. Additionally, the rotary knob in concept B presented several possible issues related to intuitiveness as there is a risk that users may be uncertain about which direction to turn the knob or how far it should be rotated.

Both the lever (C) and the button (A) are interaction methods used by Aarke in their existing carbonators to initiate the carbonation process. To maintain consistency, these two options were considered particularly relevant. Buttons are also commonly used across carbonators from other brands, further reinforcing consistency and user familiarity. The lever, in contrast, stands out more visually and physically, providing a clear affordance that it is meant to be operated. Both interaction methods for initiating carbonation, were explored further through user testing.

9.2 Carbonation Interaction Design Exploration

The exploration of the interaction design for initiating carbonation was conducted through a trigger test involving ten participants. This method was chosen since trigger materials evoke spontaneous reactions and encourage participants to rely on their own intuition, free from predefined instructions or bias. By providing a tangible and engaging stimulus, the trigger material facilitated a realistic context that supported creative thinking and authentic user responses. The primary objective was to investigate how users intuitively expect to interact with a carbonator to initiate the carbonation phase.

Test Methodology

A prototype without any predefined interaction element for carbonation was used to allow for unbiased user input. The test was structured into five distinct steps.

Step 1: Participants were asked to indicate and describe where on the prototype they would expect to interact in order to start the carbonation process. This step was completed without any guidance, relying solely on the participants' first impressions.

Step 2: Participants then described how they would prefer to interact with the point they identified in Step 1 (e.g., push, pull, turn).

Step 3: Each participant was given a lever and instructed that it was to be used for initiating carbonation. They were asked to place the lever on the prototype in the location they felt was most appropriate. In addition, they were asked to demonstrate and describe how they would like the lever to be used in the interaction.

Step 4: Participants were then provided with a round object resembling a button or knob. As in the previous step, they were asked to place it on the prototype where they believed it would be most suitable. They were also asked to demonstrate and describe how they envisioned using this element to trigger carbonation.

Step 5: The final step consisted of a discussion with the participant regarding the various interaction concepts tested, as well as their general thoughts and preferences related to the carbonation phase.

Results and Insights

Results from the test, in which the participants brainstormed their ideal functionality, indicated a preference to initiate carbonation through a button-pressing action. There was some variation in the preferred placement of this imaginary button, with suggestions including the top of the protective cover, the top of the rear column, or a smaller button located at the base of the machine. The participants expressed a preference for a downward pressing motion, as it allows for more effective use of hand strength and provides a sense of control. However, some participants also suggested placing a lever or knob on the right side of the rear column. The proposed placements and their frequency are visualized in Figure 9.9, where orange highlights the suggested areas and darker shades indicating more frequently mentioned locations.

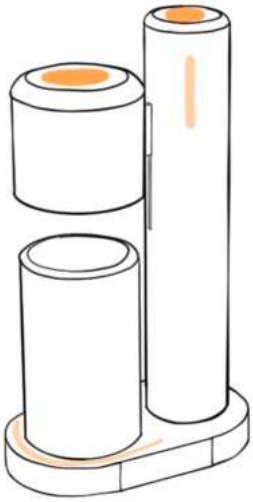


Figure 9.9: Initial placements of carbonation interaction.

When they were given a lever, most of the participants placed it on the right side of the machine, since they wanted to interact with the lever with their right hand. The most common placement was on the side of the rear column pointing forward, and the user had to push it down to carbonate. Other suggestions were to have it pointing upwards, like a beer or water tap, or downwards. All the chosen placements are shown in Figure 9.10.

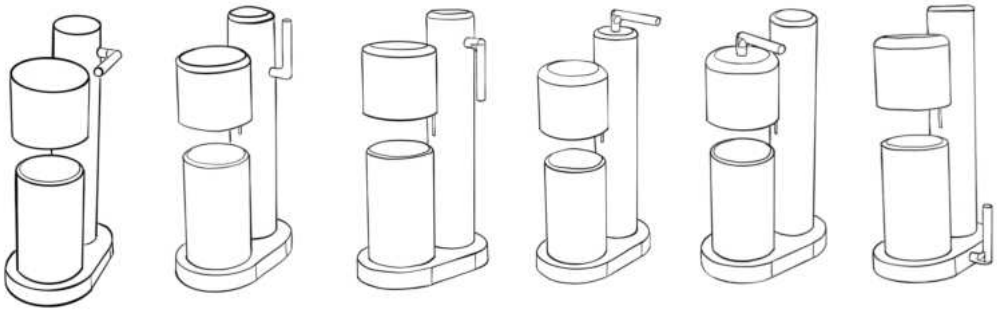


Figure 9.10: Placements of lever

Figure 9.11 shows a participant performing the test, demonstrating how the interaction was explored in practice.



Figure 9.11: Participant engaged in the interaction design test for carbonation initiation.

When interacting with the button or knob, 9 out of 10 participants preferred using the component as a push button rather than as a turning knob. One participant expressed this preference by stating:

"I want to use it like a button that you press down; it feels hard to know which way to turn it if it's a knob."

The most frequently suggested placements for the button were on top of the rear column or on top of the protective cover. Other suggestions included positioning a smaller button on the base. All mentioned positions and how the part would be used is shown in Figure 9.12.

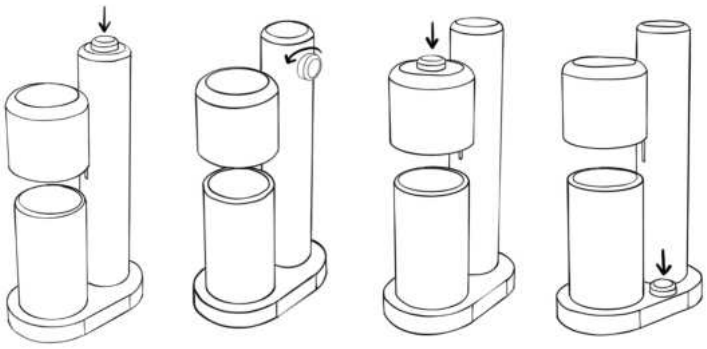


Figure 9.12: Placements of button/knob

When comparing the lever and the button after trying both, participants highlighted various advantages and disadvantages of each option:

"It is more intuitive how to interact with a lever; with a button, it is unclear whether you are supposed to press it once or hold it down, for example."

"The lever feels familiar, like a tap handle. It looks like something you naturally push downward."

"The button is more discreet and integrates better with the overall design."

When participants were asked to choose between the two options, 7 out of 10 preferred the lever. The lever was particularly favored for its intuitive interaction, as users found it easier to understand how it should be operated.

Chosen Interaction Design for Carbonation and Ventilation

The lever was ultimately selected as the preferred interaction method. As the carbonation process is the primary user interaction, the lever offers strong affordance by clearly signaling that it is meant to be grasped and moved. To support natural mapping between action and outcome, the lever was placed on the right side of the rear column, visually and mechanically linking it to the gas cylinder. Mounting the lever on the cover was intentionally avoided, as the lever should not be engaged while closing the cover. The lever is oriented forward and designed to be pressed downward during use. This downward motion is consistent with the Aarke Carbonator 3, reinforcing consistency and improving stability by minimizing the risk of the machine twisting or shifting during operation.

Various ventilation methods were considered, but it became evident that the ventilation interaction was closely dependent on the chosen method for carbonation.

Once the carbonation approach, lever actuation, was established, it naturally defined the ventilation mechanism as well. With carbonation initiated by pressing down the lever, ventilation was designed to take place as the lever is released. This principle is also applied in the Aarke Carbonator 3, where ventilation is triggered during the upward motion of the lever following carbonation. The same mechanism was chosen for the new product, as it allows both functions, carbonation and ventilation, to be handled through a single user action. This eliminates the need for an additional step or interaction point, resulting in a streamlined and intuitive experience where one movement controls two internal processes.

9.3 Smart Features Exploration

Since the carbonator is electrically powered, it opens up a wide range of possibilities for integrating smart features. The technical features considered included both safety-oriented functions and those aimed at enhancing the overall user experience.

An exploration of which smart features to integrate in the product was also done in form of a trigger material test. The purpose was to explore various additional features that could be enabled through the use of electricity in the carbonator. This test aimed to investigate what users consider to be an optimal user experience throughout the carbonation process. Specifically, it focused on identifying which functions users expect the machine to perform, and what kind of interaction or system feedback they find most intuitive and helpful at each stage. The same group of ten participants also participated in this test.

Test Methodology

The test was divided into three main steps.

Step 1: The participant was given the prototype and the different steps in the usage were described.

Step 2: Participants were encouraged to reflect on how they would prefer the machine to respond during key moments in the carbonation process, such as when the bottle is placed, when carbonation is initiated, and when the process is complete. To support this reflection, a list of seventeen potential features was provided.

For each selected feature, participants were asked to explain how and where on the machine the feature should be implemented, including the placement and type of feedback it should provide. They were also asked to describe how the feature activation or status should be indicated.

Examples of proposed features included *Indication that the CO_2 cylinder needs to be replaced* and *Indication that the protective cover is locked*. The participants were then asked to elaborate on how they imagined these features to function, including their preferred form of feedback - auditory, visual, or tactile.

Step 3: The final step consisted of a discussion in which participants reflected on the features they had selected, explained the reasoning behind their choices, and shared additional thoughts on the overall user experience and the desired forms of feedback.

Results and Insights

All suggested features and the number of participants who selected each feature is shown in Figure 9.13.

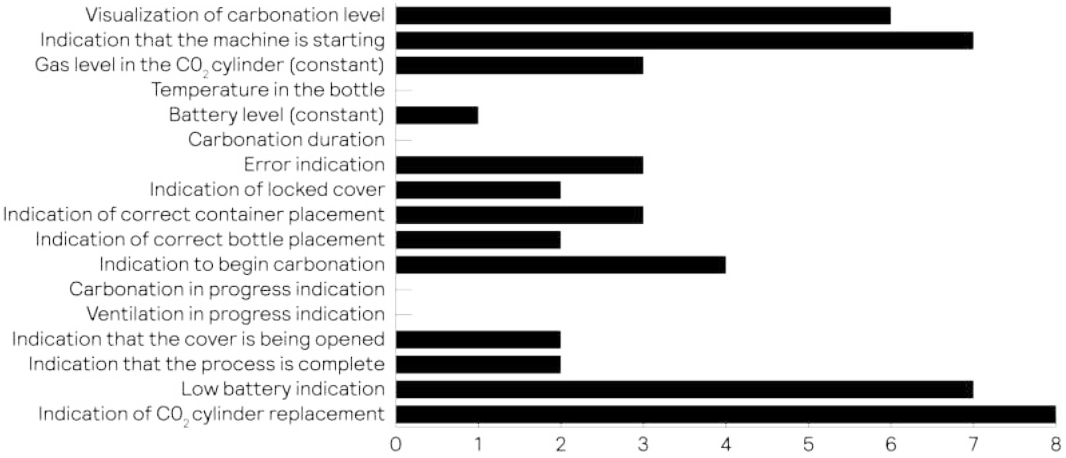


Figure 9.13: Participant preferences for the features in the trigger material test, showing the number of participants who selected each feature.

Regarding the options that did not receive votes, there was a shared perception among the participants that these features were unnecessary and did not contribute to an enhanced user experience. Specifically, the participants felt that no additional feedback was needed to indicate that carbonation was in progress, as the natural sound produced during the process was considered sufficient.

The features that a majority of the participants expressed interest in were further explored and compiled into insights based on the discussions.

Both gas and battery levels were discussed regarding whether they should be displayed constantly or only indicated when low. The majority agreed that constant visibility was not necessary. Instead, a warning was preferred when levels were

getting low, ideally early enough to allow a few more uses, giving the user time to prepare a new cylinder or charge the machine before it is completely empty. It was also commonly mentioned that these indicators do not need to be placed visibly on the front of the machine. As for the preferred method of visualization, participants unanimously favored using LED lights as the most effective and clear way to indicate status.

The visualization of carbonation level was discussed in terms of placement, form, and feedback. Common suggestions included using LED indicators or scales that light up progressively, placed either near the carbonation interface or on the front of the base. Participants highlighted the importance of clear mapping, preferring indicators positioned close to the lever or interaction point for intuitive use. Many also favored mechanical cues, such as varying lever pressure or duration, to control the level, ideally combined with a visual bar or scale. Overall, the visualization should help users feel in control while clearly signaling the current level of carbonation.

Regarding indication that the machine starts, the insights were that a dedicated indicator was not considered essential by all participants. While some suggested a diode or a green light to signal that the machine is awake and ready, others felt that this feedback could be subtle or integrated into existing elements. Several participants mentioned that the activation of another indicator or a soft startup sound could be sufficient to convey that the machine is starting. Overall, the feature was seen as helpful but not critical, and most preferred a minimal and integrated approach over a separate visual element.

General reflections were also shared regarding the features and overall design of the machine, focusing on what is intuitive and effective without overwhelming the user with too many indicators. Participants expressed concerns that an excess of LED lights or visual cues could become confusing rather than helpful. As two participants noted:

”Less is more, it can become too much. People never read manuals, so it might just be overwhelming.”

”I want a clean, minimalist machine, not too many LEDs and indicators. It should be as simple as possible for the user.”

Conclusion

A set of key electronic features was selected to improve the overall user experience. These features were chosen specifically for their ability to provide clear, timely, and

intuitive feedback to the user, aligning with Norman's principle of visibility and reducing uncertainty during the carbonation process. Rather than adding multiple new functions that could complicate the interaction, only the most essential features were prioritized. The selection was based on how effectively each feature could communicate system status and enhance user confidence in the product.

The following features were chosen for implementation:

- **Carbonation level visualization:** Provides real-time feedback on the progress of carbonation.
- **Start-up indication:** Clearly signals when the carbonator is on.
- **Low battery warning:** Alerts the user when the battery level is low and needs recharging.
- **CO_2 cylinder replacement notification:** Notifies the user when the gas cylinder is near empty, helping to avoid interrupted usage.

Chapter 10

Detailed Design

The detailed design phase marks the realization of all design decisions and selected solutions. It represents the final half of the second diamond in the Double Diamond model, where the design is fully defined and finalized. During this phase, extensive CAD modeling was carried out to develop and refine each component. The detailed design is presented in the following sections, organized by individual parts of the carbonator.

10.1 The Base

The base of the carbonator consists of two main components: an internal plastic frame that supports all internal elements and ensures structural stability, and an outer stainless steel shell that is mounted over the frame to provide a clean, durable exterior.

The overall size of the base is determined by the internal components it needs to house, particularly the electrical features. In addition to functionality, the dimensions were also chosen to convey a sense of robustness and stability. The foot's shaft angle mirrors that of the Aarke Carbonator Pro, and the decision to let part of the plastic extend beneath the metal was made to reflect Aarke's design language, seen in both the Carbonator 3 and Pro. The base's protruding section also draws inspiration from previous Aarke designs, enabling a smooth and seamless connection with the rear column. The curvature between the flat area and the protrusion follows the same radius used in Aarke's current carbonation machines. These design choices are visualized in Figure 10.1.



Figure 10.1: Side-view of the base showing shaft angle and porportions between shell and inner structural frame.

The foot houses a battery, a module that enables battery charging via a USB-C port, a PCB card, a microcontroller with a pressure sensor and a timer, and two weight sensors.

The largest component is the battery which has a designated place in the foot where it is secured by snap-fit mounts.

To enable recharging, a USB-C port was integrated into the base, allowing the user to plug in a charging cable when the battery is low. The choice to use USB-C stems from the EU regulation mandating that consumer electronics adopt this standard for charging (Regeringskansliet, 2023). The port was placed on the back, slightly to the right of the base, to keep it hidden during everyday use while still being easily accessible when it's time to plug in the charging cable, see Figure 10.2. The mounting and structural support of the port were particularly important, as the user applies pulling force to the port when disconnecting the cable after charging. The module is therefore secured with snap-fits on all sides to prevent any kind of movement.



Figure 10.2: The USB-C port in the base.

The USB-C port has been slightly recessed into the base of the carbonator. This means that when a charging cable is connected, the rigid connector sits within the outer edge of the base. As a result, the machine can still be placed close to a wall without the cable being bent or obstructed, a practical solution that protects the cable while saving space.

To control all the electrical components, a microcontroller will be used, which is also held in place with snap-fits. A pressure sensor, connected to the microcontroller, is used to measure the pressure inside the bottle during the carbonation process. To enable this, the sensor is routed from the base, through the rear column, and up to the protective cover. The base is also designed to allow a PCB board to be screwed in place, where LEDs are intended to be connected. A view of the foot, showing the intended placements of the components can be seen in Figure 10.3.

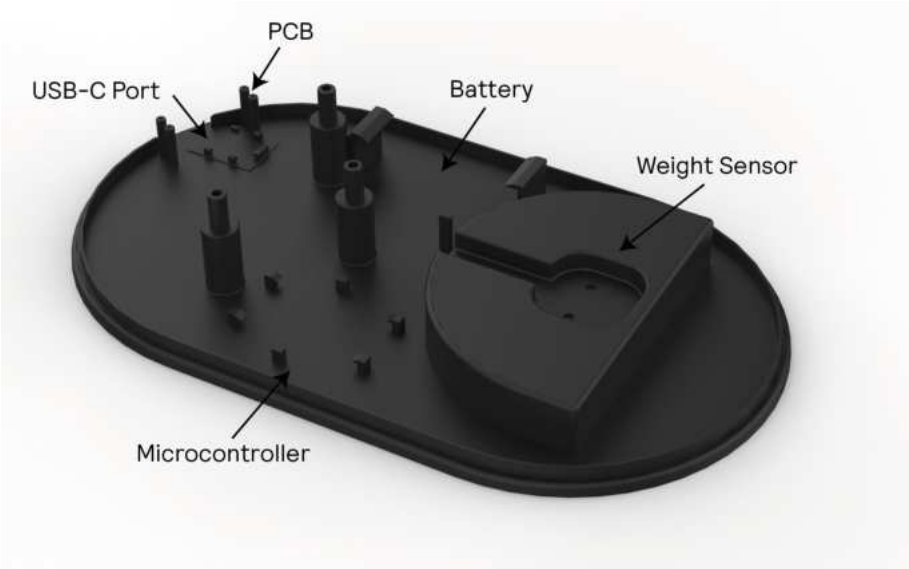


Figure 10.3: Placements of the inner components in the base.

The first weight sensor is integrated into the plastic base, positioned directly beneath the area where the container is placed. This sensor detects when a container with a filled bottle is set on the base, functioning as an automatic wake-up trigger for the machine. In this way, the carbonator exits sleep mode without the need for a dedicated on-button. This solution also prevents the user from carbonating an empty bottle, an important safety consideration.

Once a filled bottle and container are detected, the machine remains active for a maximum of five minutes. If the user does not initiate the carbonation process within that time, the carbonator automatically returns to sleep mode. If carbonation is performed during the active period, the system re-enters sleep mode one minute after the container and bottle are removed. This interaction model is inspired by digital scales that activate upon weight detection and power down after inactivity. Importantly, the sensor is calibrated to only respond to a filled bottle, allowing the user to store the container and empty bottle in place when the machine is not in use, without triggering activation or unnecessary power consumption. The sensor is calibrated to detect a weight range broad enough to accommodate slight variations in water volume, yet precise enough to ensure that both the container and a filled bottle must be present to activate the system.

The second weight sensor is placed beneath the rear column, where the CO_2 cylinder is kept. The purpose is to fulfill the feature to indicate when the cylinder needs be replaced. A plate is placed above and rests on the weight sensor, both keeping it in

place as well as protecting it. The plate can easily be lifted by the two cutouts. The CO_2 cylinder can then be placed onto the plate and thereby its weight is measured, see Figure 10.4. The sensor is active when the machine is in its awake mode. A full gas cylinder has a weight of 400-425 grams and when empty it weights 110-140 grams. Therefore the weight sensor needs to have a measurement range that covers the full weight of the CO_2 cylinder, with a resolution fine enough to detect weight changes of around 10 grams or less.



Figure 10.4: Weight sensor placement for weighing the CO_2 cylinder.

The metallic shell is mounted onto the plastic base using two flanges at the front and three screws positioned underneath the rear column. These screws fasten into an internal plastic component, clamping the shell securely in place. The screws are arranged in a triangular pattern to provide increased support and distribute the load more evenly. This configuration helps prevent a hinge-like effect that could occur with only two screws in a straight line, which might allow the shell to flex or loosen over time. See Figures 10.5 and 10.6 for details on flange placement and screw insertion points.

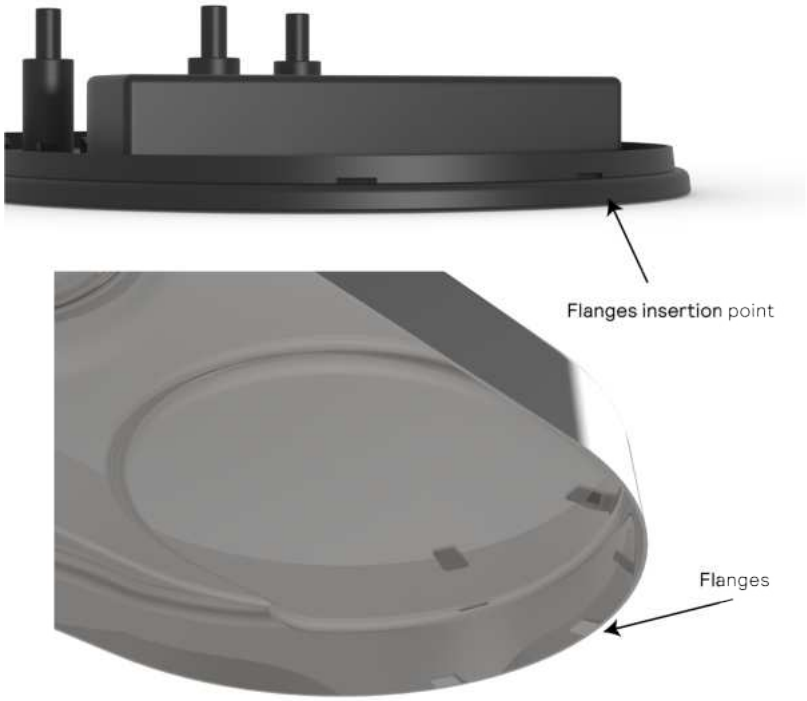


Figure 10.5: Flanges for mounting the outer foot with the inner frame.

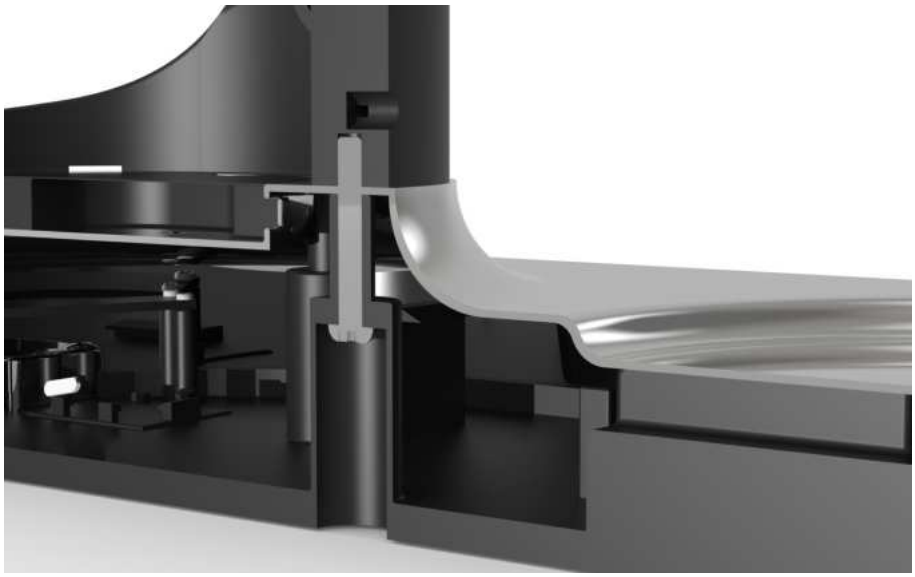


Figure 10.6: One of three screws that attach the structural base with the rear column.

Lastly, on the underside of the base, four recesses are provided for silicone feet, see Figure 10.7, preventing the machine from sliding or moving during use.



Figure 10.7: Silicone feet on the underside of the base.

10.2 The Container

Regarding the container, the shape was already set since it should align with the already existing bottle, and also resemble a cooler to indicate that usage area. However, dimensions, such as heights, diameters, and roundings was iterated on in CAD, in order to make prototypes to test the various dimensions, see Figure 10.8.



Figure 10.8: Iterations of container dimensions.

A relatively small outer diameter was preferred to allow the user to grip the container with one hand. Additionally, a tight fit between the bottle and container was chosen to reduce the risk of misalignment and to minimize noise caused by contact between the glass bottle and the stainless steel container. To achieve this, a diameter difference of 3 mm between the bottle and the inner wall of the container was designed, ensuring a snug yet functional fit. This dimension was verified by testing with a 3D-printed prototype of the container together with a physical bottle, confirming that the fit allowed for easy placement while providing the intended stability.

The height of the container was determined based on two key considerations: ease of bottle handling and its relationship to the protective cover. To ensure that the user easily could place and remove the bottle, the container could not be too tall, as this would make it difficult to grip the bottle once inserted. In contrast, a shorter container would require a taller protective cover, as more of the bottle would need to be encapsulated from above. A taller container, on the other hand, would already shield a greater portion of the bottle, allowing for a more compact cover. To determine the optimal height, three variations were tested, spanning a total range of 2 cm between the shortest and tallest. In the end, the medium height was selected, striking a compromise between usability and the desired proportions of the protective cover.

The container features a double-walled design with an air-filled gap between the inner and outer walls, allowing it to function as a cooler. To accommodate this, the outer diameter is 20 mm larger than the inner diameter, adding 10 mm in radius on each side. Despite the added insulation, the outer diameter is kept as small as possible to ensure the container remains easy and comfortable to grip with one hand.

For the roundings, a small radius was chosen at the top of the container to ensure a mostly straight and clean interface between the protective cover and the container when the cover is in its lowered position. This design choice helps maintain a compact and seamless fit in the assembled state. In contrast, a larger rounding was applied at the base to provide a smooth transition to the surface below.

A prototype demonstrating the fit between the container and bottle is shown in Figure 10.9.



Figure 10.9: The bottle being placed into the container prototype.

To further address the sound that occur between the bottle and steel container, a silicone ring was integrated at the bottom of the container. This element cushions the bottle, ensuring smoother placement and significantly reducing the metallic sound that could otherwise occur when glass meets steel. A section view of the container highlighting the silicone ring is shown in Figure 10.10.



Figure 10.10: Section view of container with silicone ring.

10.3 The Protective Cover

The design of the protective cover was shaped by the relationship between the container, the bottle, and the internal components. It is composed of several parts, each serving a specific function. At its core, a plastic structural frame supports the internal components, including the nozzle and tubing. As many of the internal components will be similar to those used in the Carbonator 3, these parts will not be further discussed in detail in this report. The components and the plastic frame is then enclosed in two stainless steel shells, an inner and an outer, forming the final exterior of the part. A sectional view of the cover illustrating these components is presented in Figure 10.11 where the white part represent the inner components.



Figure 10.11: Section view of the protective cover.

The outer shell, provides the visual and structural alignment with the container's exterior surface and is the part that the user sees. It also works as protection for the internal components. The outer diameter of the shell was defined to ensure that, when lowered into its operational position, the cover aligns seamlessly with the container, creating a clean visual transition and secure physical fit. The height was determined by the need to accommodate both the bottle and the internal components required for carbonation. Since creating detailed CAD models of these components was outside the scope of the project, feasibility was assessed by comparing the available volume with that of the Aarke Carbonator 3. It was reasoned that if the Carbonator 3's head and its internal elements could fit within the new outer shell, the current configuration would be sufficiently sized and therefore viable for the electric model as well. A cut-out was added to the outer shell to enable an opening for the bridge that connects the protective cover to the rear column.

The inner shell of the cover is designed to align with the container's interior when the cover is lowered. This shell also serves as internal protection, shielding components from water and enclosing the carbonation environment in stainless steel during operation. The connection between the cover, bottle, and container is illustrated in Figure 10.12.

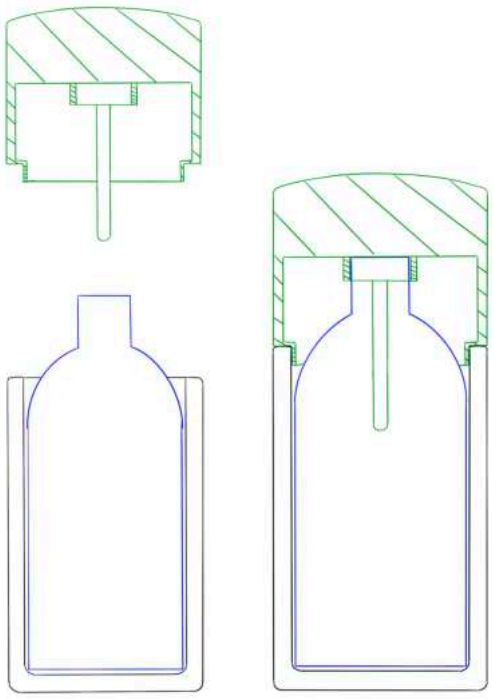


Figure 10.12: Section view illustrating the connection between the cover, the container, and the bottle.

The plastic component inside the cover serves as a structural frame for the internal elements. It features a circular ring that extends through the inner shell, securely enclosing the mouth of the bottle to ensure proper alignment. A detailed construction of the sealing mechanism between the bottle and the plastic component will not be presented in this project, as it is intended to function in the same way as in Carbonator Pro. The plastic frame also features the connection point for attaching to the part bridging between the protective cover and the rear column.

The ventilating air from the ventilation process, when the lever is released, is released into the container. To accommodate this, a small cutout was made to allow for the ventilated air to be safely released, see Figure 10.13. This is further explained in Section 10.5.



Figure 10.13: View from underneath the protective cover showing the cutout for ventilated air.

The internal components, inner plastic frame, and inner shell are assembled using four screws, which are inserted from the inside of the inner shell. These screw placements are visible in Figure 10.11. Once this assembly is complete, the outer shell is attached on top using a press-fit connection.

10.4 Bridge between cover and rear column

A bridge connects the protective cover to the rear column, enabling its vertical movement, housing the locking mechanism, and providing a pathway for the tubes routed from the back of the carbonator into the cover. It is mounted to the cover with screws connecting it to the inner plastic structure and on the side facing the rear column, the bridge is shaped like a T and fits into a corresponding T-track within the rear column. This T-track guides the vertical movement of the protective cover, with its length calibrated so that the bridge reaches the end of the track precisely when the cover is in its fully lowered position.

The locking mechanism uses a pull-type solenoid equipped with a spring. During the downward motion, the spring presses the solenoid pin against the inner surface of the T-track. At the bottom end of the track, a hole allows the pin to engage and lock the vertical movement of the cover. Once carbonation and subsequent ventilation are complete, the solenoid is activated to disengage the lock. A pressure sensor monitors

the internal pressure of the bottle, and when the pressure drops to a predefined threshold, a signal is sent to the solenoid. This causes the pin to retract from the locking hole, re-enabling vertical movement and allowing the cover to be lifted. The solenoid is held in place by a plastic holder inside the bridge.



Figure 10.14: The bridge and its housing for the solenoid.

The user receives feedback during this stage through the audible sound generated by the solenoid. A distinct clicking sound is produced both when the solenoid is engaged to lock and when it retracts to unlock. This sound provides confirmation that the locking mechanism is functioning correctly and signals to the user that the cover is about to open. The exact sound depends on the specific solenoid.

10.5 The Rear Column

The rear column is a cylindrical component that serves four main functions: housing the CO_2 cylinder, linking the external lever to the gas valve to release CO_2, enabling the vertical movement of the protective cover, and providing structural support for the entire machine. In line with the rest of the product construction, the rear column consists of an internal structural frame made of plastic and an external, stainless steel shell which in turn is divided in to three parts, a front-, a back- and a top-part. The design and construction of this part were largely influenced by the user's access to the CO_2 cylinder during replacement.

To accommodate internal components placed at the top of the column, top-loading of the gas cylinder was deemed unfeasible. Consequently, the CO_2 cylinder is designed

to be accessed and replaced from the back of the machine. To enable this, the rear shell was divided into two symmetrical halves. This split provided sufficient space for the user to remove and insert the cylinder with ease.

For the opening mechanism of the rear shell, inspiration was drawn from existing Aarke products. A spring-loaded button was selected as the user interaction point, specifically the same design as the button used to open the lid of the Aarke Kettle, see Figure 10.15. By using the same type of button, the design language is reinforced, a clear connection to the Aarke brand is established, and a sense of familiarity is provided for the user.



Figure 10.15: Button to back of rear column to access the gas cylinder.

To open the back part of the rear column, the user presses the spring-loaded button, which in its default position covers the seam between the upper and lower parts of the shell. Internally, the button includes a small latch that hooks into the upper section of the shell. Pressing the button releases this latch, allowing the rear shell to tilt outward. The button is designed to provide tactile feedback, with a slight resistance that must be overcome. When the user reaches the bottom of the button's travel, a distinct feeling confirms that the latch has been disengaged and the rear panel can be pulled open. A section view of the spring-loaded button is visualized in Figure 10.16.

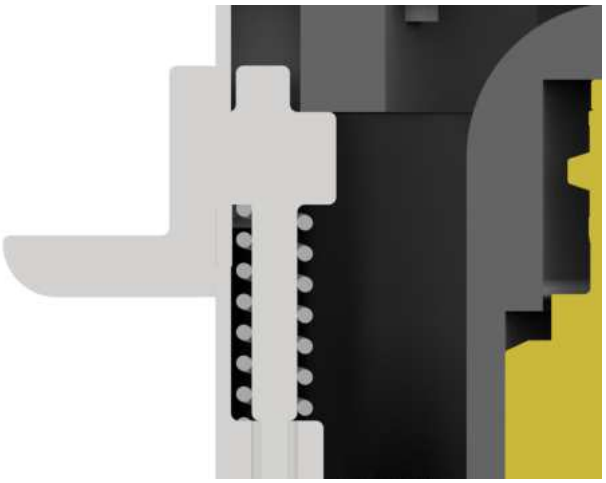


Figure 10.16: Section view of rear column with the spring-loaded button used to open the rear shell.

At the bottom of the rear shell, two flanges are inserted into the internal frame to ensure secure closure and alignment. The flanges are shown in Figure 10.17.

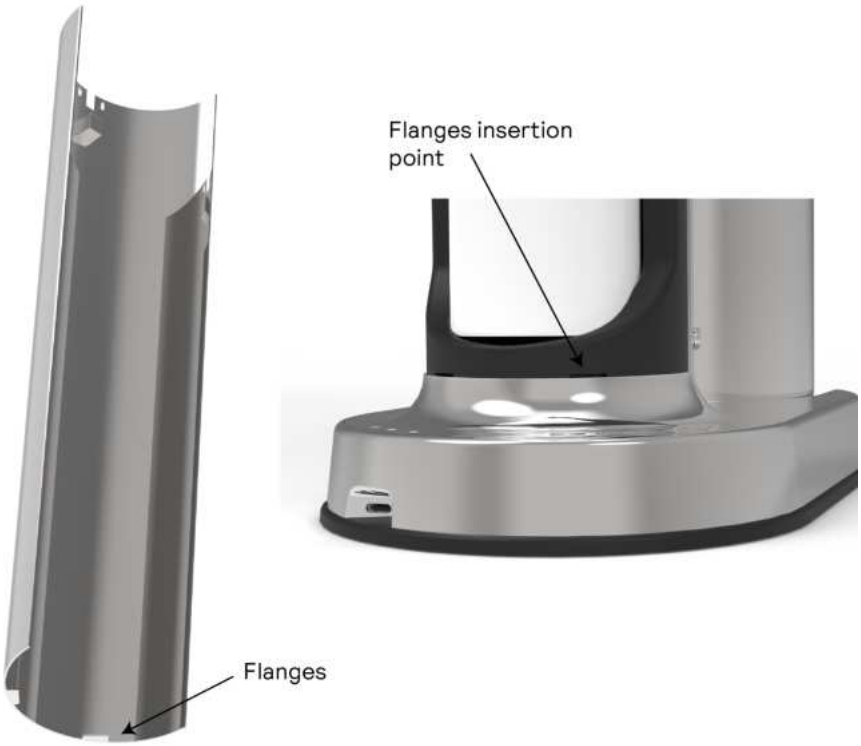


Figure 10.17: Flanges for mounting the rear shell with the inner structural frame.

Once the shell is removed, the CO_2 cylinder can be lifted vertically off the platform it rests on, guided by a T-slot mechanism. Vertical movement is necessary to lift the cylinder over a plastic threshold and to allow it to be weighed. Since the cylinder needs to be both screwed into place as well as be resting on the weighing plate, vertical movement, enabled by a T-slot mechanism was the simplest solution to meet both requirements. The T-slot is shown in Figure 10.18.



Figure 10.18: The T-slot of the CO_2 cylinder (here white) and the inner structure of the rear column to allow for vertical movement.

To facilitate the screwing of the cylinder into place, the design also allows the user to tilt it outward. This is achieved by mounting the part into which the cylinder is screwed onto an axle, allowing it to rotate, see Figure 10.19. The holder follows the shape of the gas cylinder to guide the user in how and where the cylinder is supposed to be placed.



Figure 10.19: Axle that allows for the CO_2 cylinder and its holder to tilt.

To make it easier to grip the cylinder, cutouts have been made in the inner plastic structure, allowing the user to insert their fingers and get a better grip around the cylinder, see Figure 10.20.



Figure 10.20: Cut-out section in inner structural frame allowing the user to reach and grab the CO_2 cylinder.

The inner structural frame provides support for the rear column, helps with guiding cords from the foot to the top and works as a base to which the other components can be attached. To enable access to the CO_2 cylinder, the frame is open at the rear-facing side of the machine. However, for added stability, the frame remains fully circular at the bottom. The structural frame is mounted on the foot from underneath the base with three screws.

Integrated into the structural frame are also a T-track, which guide the vertical movement of the protective cover, see Figure 10.21.

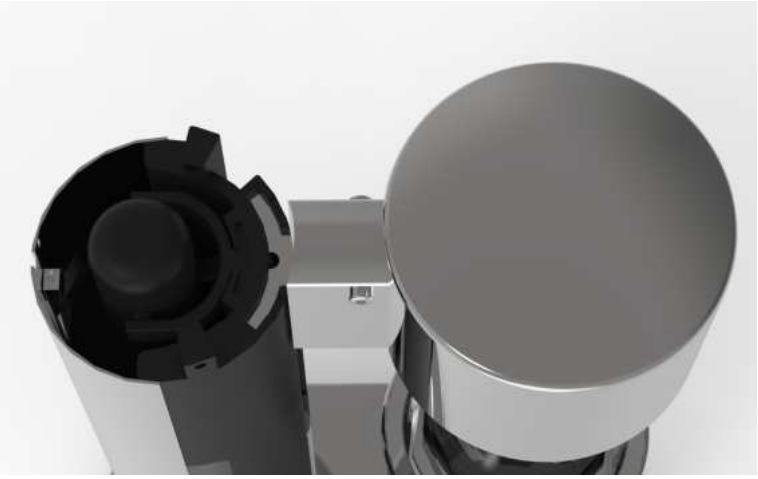


Figure 10.21: The T-track of the protective cover and the inner structure of the rear column to allow for vertical movement.

The motion of the protective cover is supported by a gas spring, mounted with one end inside the rear column and the other on the underside of the bridge. This gas spring ensures a controlled and smooth movement, particularly during the opening sequence of the cover.

The other half of the outer shell is mounted to the inner frame by screws at the bottom. The shell has a cutout that aligns with the T-track for the bridge which is shown in Figure 10.22.



Figure 10.22: The front stainless steel shell of the rear column. Cutouts of screws and LED at the bottom and a track for vertical movement of cover.

The two flanges at the top of the outer shell are designed to be secured between the inner structure and the top section of the rear column. This method allows the top of the outer shell to be fastened without the use of visible screws, contributing to a cleaner exterior appearance.

The upper part of the rear column houses the core components responsible for carbonation and ventilation. Internally, this section contains various valves, one set to initiate carbonation and another to release pressure for ventilation. These are controlled mechanically via a lever mounted directly on the upper rear column, positioned close to the components it operates. When the lever is pressed down, the gas valves open to allow CO_2 flow; when released, the ventilation valves activate to safely release pressure.

To manage the airflow during ventilation, the released gas is directed into the container, an approach also used in the Aarke Carbonator Pro. This is achieved by routing a ventilation tube from the rear column into the cover. An additional tube connects the CO_2 cylinder to the nozzle, enabling controlled gas delivery during carbonation.

Structurally, the upper part of the product consists of a plastic internal frame and a stainless steel outer shell. The plastic frame supports the upper end of the CO_2

cylinder and secures the internal components in place. It is mounted to the rear column's inner structure, to which the stainless steel shell is attached. The shell serves both protective and aesthetic purposes and is fastened to the frame using two screws.

10.6 Smart Features and User Experience Interface

The interface of the carbonator's user experience (UX) design was developed through iterative prototyping and testing, with a primary focus on achieving an intuitive and user-friendly interaction. As part of the interface, visual indicators were integrated to support the selected smart features.

Battery level visualization was implemented using a set of LEDs. The placement of these indicators was guided by the principle of mapping, ensuring that visual cues are located in close proximity to the features they represent. Consequently, positioning near the charging port was explored. Various placements were evaluated both in CAD and on physical prototypes, including on the side of the base, on top of the base, and within the surface surrounding the charging port.

Testing revealed that positioning the indicators on the top surface of the base provided the best visibility for the user. This location allowed for discreet integration that maintained a minimalistic appearance while still offering sufficient visibility when needed. This decision also aligns with findings from earlier user testing during the smart feature exploration phase, which showed that users did not require constant visibility of the battery status.

The battery indicator consists of three LEDs that gradually turn off as the battery level decreases. This setup was chosen to provide the user with a clearer sense of the remaining charge, which would be harder to achieve with a single low-battery warning light. Three LEDs were considered sufficient to convey this information without adding unnecessary complexity or visual clutter. An illustration of the battery indicators can be seen in Figure 10.23 and the final placement of the LEDs are presented in Figure 10.24.

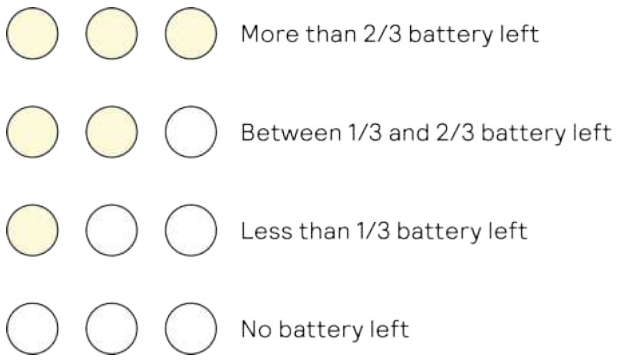


Figure 10.23: Battery indicators.



Figure 10.24: Placements of indicators for battery level.

An LED was also chosen as the indication method for when the CO_2 cylinder needs to be replaced. In this case, the placement was also intended to be connected to and close to the gas cylinder. As a result, it was positioned on the rear column. Various positions were tested in the CAD model before finalizing the location.

The LED indicator was positioned at the bottom of the rear column on the front of the machine, drawing inspiration from the Aarke Coffee Maker. Although the LED in the Coffee Maker serves a different function, this was not considered an issue, as single LEDs are widely used across various products to signal different statuses. In this case, the LED flashes red to indicate a low gas level and shifts to a continuous red light when the gas is completely depleted. This approach is similar to the feedback system used in electric toothbrushes, where a flashing light signals a low battery. Once the flashing begins, the user has approximately three carbonation

cycles remaining. A single LED was deemed sufficient for this purpose, as users also receive additional, intuitive feedback during use, such as a noticeable increase in carbonation time when gas levels are running low.

The placement of the LED is presented in Figure 10.25.



Figure 10.25: Placements of indicator for gas level in CO_2 cylinder.

Due to its placement, the LED is only visible to the user before placing the container on the base and after removing it following the carbonation process. As a result, the indicator is not visible during carbonation, reinforcing that it is unrelated to the active carbonation sequence. This design choice aligns with user testing insights, which indicated that constant visibility of this indicator was not necessary.

To visualize the level of carbonation for the user, LEDs were additionally selected as the indicating method. Since the actual gas content in the water depends on several variables, such as water temperature and amount of water, it is not feasible to present an exact measurement of carbonation in real time. Instead, this product estimates the carbonation level using two key parameters: the internal pressure within the bottle and the duration for which the gas valve remains open. These factors are used to provide an approximate indication of how much gas has been added to the water.

Both the pressure sensor and the timer are implemented in the microcontroller. The pressure sensor is connected to the protective cover and provides real-time pressure data. The timer starts when the gas valve opens and stops when it closes. A signal from the lever mechanism at the top of the rear column is sent to the microcontroller to indicate when the gas valve is active.

Regarding functionality, two alternatives were considered: either having the LEDs illuminate sequentially as the carbonation level increased, or using a flowing bar-like animation to represent the gas level dynamically.

The sequential illumination method was ultimately chosen, where each LED corresponds to a distinct level of carbonation: one for low, two for medium, and three for maximum carbonation. This approach was selected for its clarity and ease of interpretation, allowing users to quickly understand the carbonation level without ambiguity. It also aligns with the overall minimalistic design language of the product, while still offering meaningful feedback during use.

Regarding placement, the same principle was used as with the previous indicators, prioritizing high visibility and mapping. These LEDs are the only indicators directly connected to the carbonation itself, making it essential that users can clearly see them while operating the machine. The upper section of the rear column was identified as the most suitable area for placement, given its proximity to the lever mechanism and its natural visibility during use. Both the upper surface and the front face of the column were evaluated as potential placements.

Due to the height of the rear column, the front face was chosen to ensure visibility even when the product is placed on a higher surface, such as a kitchen shelf. The LEDs were positioned slightly to the right of center, aligning more clearly with the lever mechanism and reinforcing the visual connection between user action and feedback. The indicators were arranged vertically and illuminate sequentially from bottom to top. This upward progression was considered intuitive, as it aligns with common conventions for increasing levels (e.g., volume bars or battery indicators), making it easier for users to quickly interpret the carbonation level. The chosen placement is shown in Figure 10.26.



Figure 10.26: Placements of indicator for gas level in the bottle during carbonation.

The indicators for battery level and carbonation level were designed to use a neutral white light, creating a clean and consistent interface that aligns with the LED aesthetics of the Aarke Kettle. In contrast, the gas level indicator was assigned a red light to convey a warning signal, as it is represented by a single LED. This use of red enhances visibility and urgency, helping to clearly communicate when the gas cylinder requires attention.

To indicate that the carbonator wakes up from its sleep mode, once a filled bottle and container are placed on the machine, the LEDs for battery and carbonation level light up.

Chapter 11

Final Design

The final design of the Aarke Carbonator E-Pro is presented in this chapter. It includes all the functions, user steps, how the carbonator is assembled, main measurements, and material choices.

11.1 Aarke Carbonator E-Pro

The Aarke Carbonator E-Pro is a rechargeable electric carbonator designed to deliver a seamless and intuitive user experience.

As Aarke's first venture into the category of electric and smart carbonators, the E-Pro marks an important step in expanding the brand's product portfolio and exploring new opportunities within the home appliance market. By combining smart functionality with intuitive mechanical interaction, the product offers a distinct user experience that preserves Aarke's signature simplicity while introducing new layers of convenience and feedback.



Figure 11.1: Aarke Carbonator E-Pro

Beyond the interaction design, the Aarke Carbonator E-Pro is designed to visually align with Aarke's established brand identity and design language. Key elements such as the domed top surfaces of both the protective cover and the rear column reflect signature Aarke geometry. Visible parting lines between components, along with the use of consistent, clean shapes and proportions, further reinforce the brand's cohesive and minimalistic aesthetic, see Figure 11.2.



Figure 11.2: Close up of domes and parting lines

The removable container, which also functions as a cooler for the bottle, contributes to a compact overall design. After the carbonation process, the user can remove both the container and the bottle and place them directly on the table to keep the water cool for a longer period, see Figure 11.3.

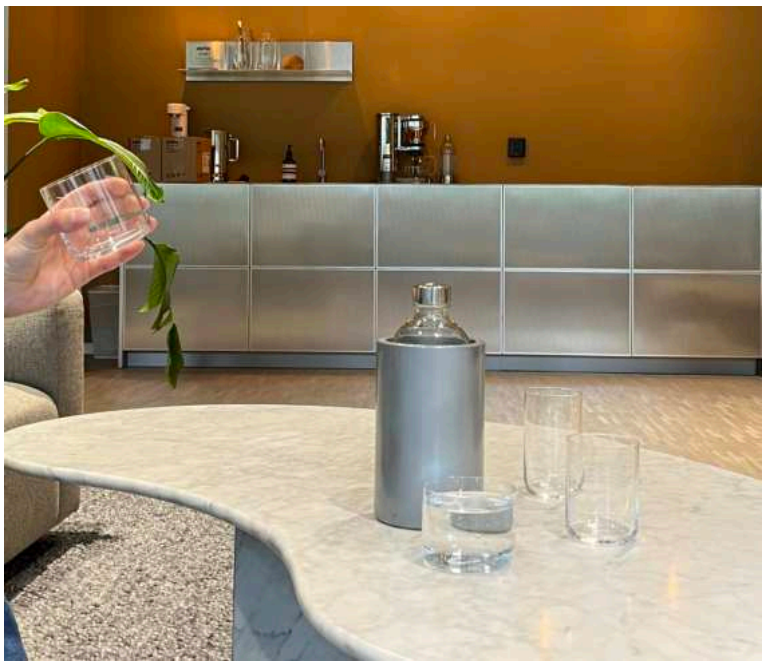


Figure 11.3: The container functioning as a cooler.

With the introduction of the Aarke Carbonator E-Pro, Aarke now offers two carbonators designed for glass bottles and one for plastic bottles. Each model brings unique qualities to cater to a variety of user preferences. What unites them is a shared focus on high quality, user safety, and a minimalistic, premium design aesthetic. All three models are shown in Figure X.



Figure 11.4: Aarke Carbonator Pro, E-Pro, and 3.

User Interaction

To carbonate a bottle of water with the Aarke Carbonator E-Pro, the user first fills the bottle and places it into the container. Once properly positioned, the battery and carbonation level LEDs light up to indicate that the machine is powered on.

To fully enclose the bottle, the user presses down on the top of the protective cover with one hand until it locks into place at its lowest position, securely sealing the bottle inside. When the lock engages, the solenoid can be heard activating with a click.

Once the bottle is enclosed, the user initiates the carbonation process by pressing down the lever. As gas flows into the water, LEDs light up sequentially to indicate the

increasing carbonation level. Auditory feedback is also provided when the maximum carbonation level is reached.

When the user releases the lever, the system automatically begins ventilation. The protective cover then opens back to its initial position once ventilation is complete. The user can now remove the container and bottle from the carbonator, completing the carbonation process.

The entire process is visualized in Figure 11.5.

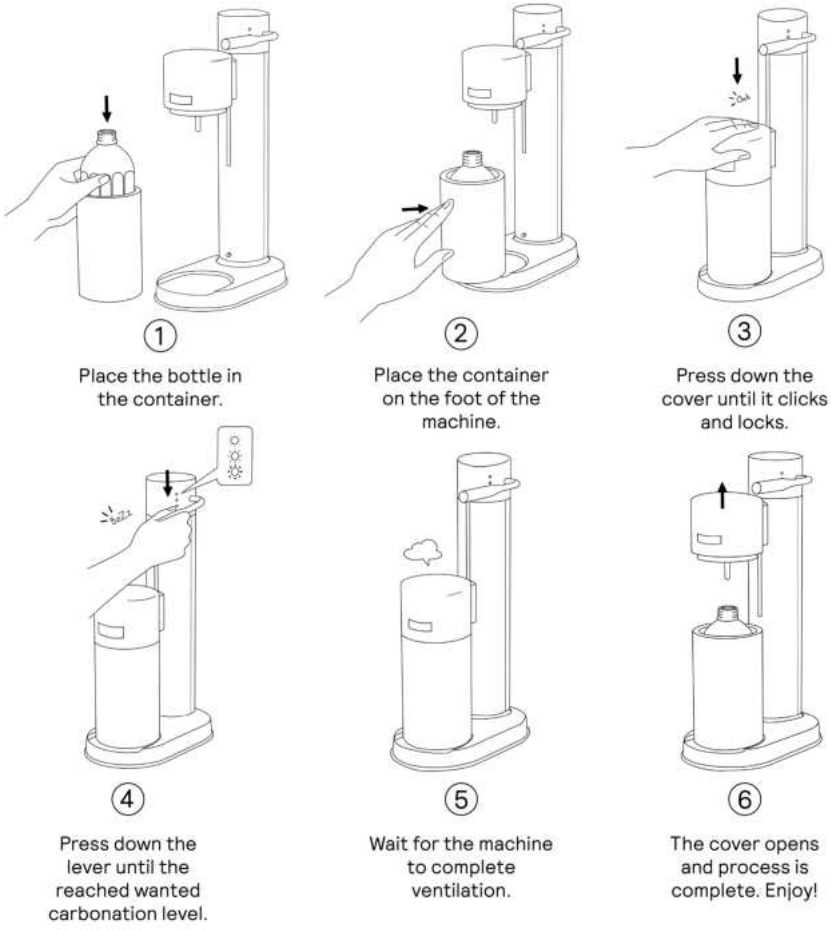


Figure 11.5: Interaction steps of the Aarke Carbonator E-Pro.

A red LED alerts the user when the CO_2 cylinder needs to be replaced. It flashes to indicate that the gas level is low and remains steadily lit when the cylinder is empty.

To replace the CO_2 cylinder on the Aarke Carbonator E-Pro, the user begins by opening the rear cover. This is done by pressing down the button on the back of the machine, which releases the cover. With the back open, the user can with ease access the empty CO_2 cylinder. Simply lifts the cylinder off the weighing plate and tilts it outward to unscrew it from the holder, see Figure 11.6. Once the old cylinder is removed, the new CO_2 cylinder is placed into the holder and screwed into position, ensuring a secure fit.



Figure 11.6: The gas cylinder in its vertical and tilted position with the back shell removed.

After the new cylinder is in place, the user closes the rear cover, ensuring it clicks into position and locks securely. This process allows the machine to be ready for use with a fresh CO_2 cylinder without the hassle of having to turn the entire machine upside down.

The process of replacing the CO_2 cylinder is presented in Figure 11.7.

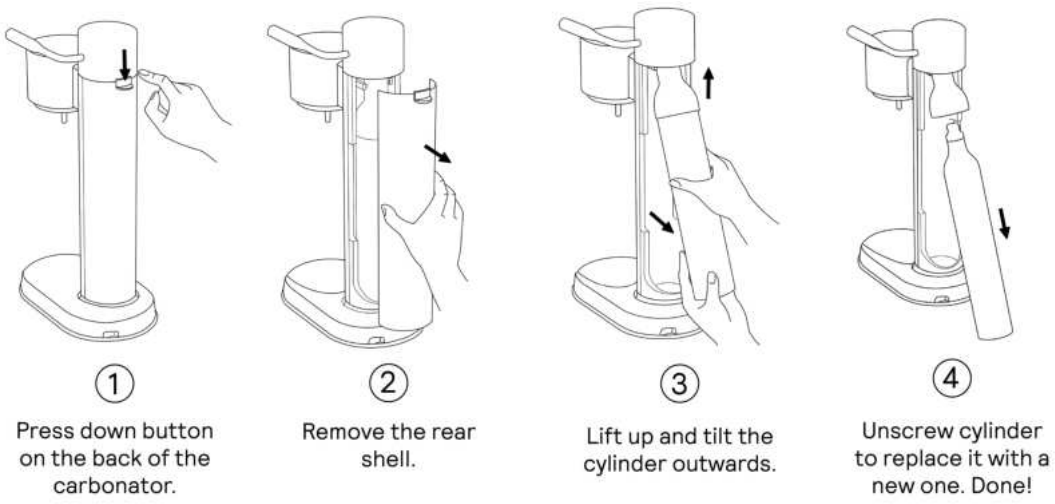


Figure 11.7: Process of replacing the CO_2 cylinder.

Assembly

The Carbonator E-Pro is designed for easy assembly and disassembly, making it simple to replace parts if something breaks or needs maintenance. All components can be taken apart without damaging any part of the machine. Below, in Figure 11.8, is a cross-sectional view of the entire machine.



Figure 11.8: Cross-sectional view of the machine

Assembly begins with the plastic base component, where all electronic parts are mounted. Once the components are in place, the metal outer shell of the base is attached, making sure that any necessary cables are routed through the designated opening. The assembled base is then set aside.

Next, the lower end of the gas spring is fastened to the internal structural plastic part with two screws. The bridge and protective cover are assembled separately and then slid into the T-slot, where they are connected to the gas spring.

Cables are routed through the designated channels in the structural plastic part, after which the front metal shell is slid on from underneath and secured with screws. The assembled components are then placed onto the foot and fastened with screws from below.

The top section is assembled separately and then mounted onto the structure with screws. Finally, the rear metal shell is attached with the help of its flanges and button, the assembly is now complete. An exploded view of the carbonator is shown in Figure 11.9.



Figure 11.9: Exploded view of the machine

Measurements

The main measurements of the product, and a size comparison between the Carbonator E-Pro, Carbonator 3, and Carbonator Pro can be seen in Figure 11.10.

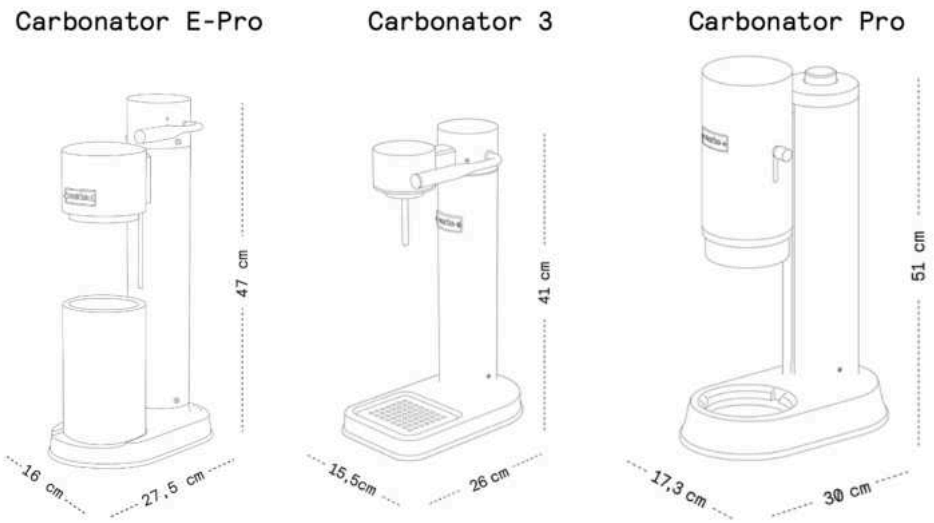


Figure 11.10: Main measurements of the Carbonator E-Pro compared to Carbonator 3 and Carbonator Pro.

The Carbonator E-Pro is more compact than the existing Aarke Carbonator Pro in every dimension, measuring 47 cm in height compared to the Carbonator Pro's 51 cm. This reduction makes the E-Pro more suitable for a wider range of kitchen environments. In typical kitchen design, a distance of 50 cm is recommended between the countertop and the overhead cabinets (**counter**). With its lower height, the E-Pro comfortably fits within this space, allowing it to be placed freely under most cabinets, compared to the Carbonator Pro, which exceeds this limit.

Material

The main materials utilized in the carbonator is BPA-free plastic and stainless steel, the same materials that Aarke has in their other models.

Aarke Carbonator E-Pro comes in three colors; stainless steel, matte black, and matte sand. The stainless steel comes in a polished finish while the black and sand are matte and textured.



Figure 11.11: Carbonator E-Pro in the three colors.

Chapter 12

Discussion

This discussion chapter explores the main results of the project, evaluating both the design solutions and the development process. It also reflects on key takeaways for both the project team and Aarke, highlighting valuable insights and lessons learned. Finally, it outlines suggestions for future work and possible directions to further develop and refine the product and design approach.

12.1 Reflection of Project Outcome

When comparing the final concept to the original Carbonator Pro, several important improvements emerge, alongside some trade-offs that need to be considered in future development.

One change is the removal of the locking mechanism. In the original product, this component was frequently mentioned in user feedback as a source of confusion and frustration. By eliminating it entirely in the new design and replacing it with an electrical solution, we have not only improved the usability but also reduced the mechanical complexity. However, the gas spring remains, meaning that the current problems with the vertical movement may still exist.

In general, the interaction flow has been greatly simplified. We merged several interaction steps into fewer, more intuitive actions, which has led to a product that we hope is easier to understand and use. For instance, we removed the need for users to manually vent the system and open the protective cover, two steps that, according to our user studies, were particularly error-prone and confusing. The new concept automates these interactions into a seamless sequence, reducing cognitive load and creating a smoother experience.

Another clear improvement is the overall compactness of the design. The final concept is smaller and takes up less counter space, which directly addresses one of the key criticisms of the original Carbonator Pro, its large footprint, especially in height. This makes the product easier to fit in typical kitchen environments, aligning with everyday usage scenarios.

From a brand consistency perspective, the new concept also aligns more closely with the Carbonator 3. By adopting the same carbonation method, a lever, and streamlining the interaction pattern, we create a more unified product family. This consistency could enhance the overall user experience for returning customers and strengthen Aarke's brand identity through a more coherent product ecosystem.

We have also introduced multifunctional components to increase the perceived value of the product. For example, the bottle enclosure system now doubles as a cooler, offering users a "delighter" feature, an added benefit that is not strictly necessary for function but adds enjoyment. This not only enhances emotional engagement but could serve as a differentiating factor on the market.

However, these improvements come with some trade-offs. The most obvious is the introduction of electronic components. While they allow for better feedback and interaction design, they also add complexity, require power, and may impact the product's longevity and repairability. The original Carbonator Pro has the advantage of being fully mechanical, an aspect some users appreciate for its reliability, independence from power sources, and reduced risk of technical failure over time. Moreover, although we have not conducted a full cost analysis, it is reasonable to assume that the addition of electronics will increase production costs, which in turn will raise the retail price. This could affect accessibility and shift the target market further into a luxury segment.

In summary, the new concept demonstrates improvements in usability, compactness, and product experience. It offers a more intuitive, elegant interaction flow and better aligns with both user expectations and Aarke's broader product line. At the same time, it introduces new challenges, particularly in terms of complexity, cost, and potential maintenance requirements. Whether these trade-offs are acceptable will depend on the strategic direction of the brand and how future development addresses these concerns.

12.2 Reflection on Project Approach

Looking back at the past weeks and the completed project, we are proud to say that we are very satisfied with the result. We both dedicated a great deal of time and energy to developing what we believe is an improved version of the Carbonator Pro.

The structure of the project followed a format similar to many previous projects at KTH, but this was the first time we could devote 100 percent of our time to a single task. It was both enjoyable and rewarding to experience what it's like to fully focus on one project. We discovered how much more can be accomplished when our attention isn't split between multiple assignments.

We're particularly pleased with how we managed our time, frontloading the project with extensive research and information gathering proved highly beneficial during the later stages. That said, we also realized how much work is truly involved in developing a new product. Despite working full-time on this, there's still a lot left to do before it could be considered a finished, market-ready product.

In hindsight, we may have set slightly too ambitious goals at the start. Narrowing the scope earlier, especially regarding the level of detail in the design phase, might have been a better strategy. However, solving all the small challenges within the detailed design process turned out to be one of the most fun and rewarding aspects of the project.

We worked very well together, balancing each other's strengths and weaknesses, which allowed us to complete the project smoothly. One of the most unexpected turns was our decision, quite late in the process, to switch direction and develop an electric version of the product. This shift required us to rethink our deliverables and invest a lot of time in researching electronic components. Despite the added workload, we believe it was the right decision and one that ultimately strengthened the project.

One major insight we take with us is the importance of interaction design. When done well, it goes unnoticed, but when it is poor, it becomes glaringly obvious.

12.3 Integration of User Observations in Product Development Projects

Aarke is renowned for its strong emphasis on mechanical engineering, aesthetics, and safety, but traditionally has not prioritized interaction design and user experience to the extent explored in this project. Their development process has mainly focused

on internal testing, without routinely incorporating user observations or usability testing as standard practice.

In contrast, this project has adopted a user-centered approach, placing interaction design and usability at the core. By involving real users through observations and testing, it uncovered concrete challenges and valuable design opportunities that might otherwise have been overlooked.

Integrating a comprehensive usability phase into the design development process requires significant time and resources, especially when multiple iterations of user testing are involved. This is an important consideration when introducing such methods into an established development framework.

This work serves as an initial step and potential inspiration for how Aarke might incorporate user testing and interaction-focused methodologies in future projects. It demonstrates how early user insights can lead to more intuitive and user-friendly product experiences, while still aligning with Aarke's core design values.

12.4 Electric vs. Mechanical Carbonators

A central consideration throughout the project has been whether the carbonator should operate electrically or mechanically. Each method presents unique advantages and limitations, influencing not only the product's functionality but also its perceived value and user interaction.

Mechanical carbonators, such as Aarke's current product range, offer a high degree of reliability, simplicity, and freedom of placement. They do not require a power source, which makes them more portable and reduces complexity in both design and usage. The manual operation reinforces a sense of control and craftsmanship that aligns with Aarke's premium and minimalist identity. Additionally, mechanical designs often have fewer failure points and are easier to maintain over long periods.

However, mechanical systems have limitations in terms of precision, adaptability, and user feedback. Without digital components, it becomes more difficult to integrate features like status indicators, usage tracking, or automated safety checks.

Electric carbonators, on the other hand, offer enhanced functionality. They can automate the carbonation process, provide clearer feedback through lights, sounds, or screens, and potentially adjust carbonation levels more precisely. From a usability standpoint, they could reduce user error and make the process easier and more automated.

The downsides of electric systems include increased complexity, higher production cost, potential maintenance issues, and the need for charging or being near a power source. These factors may reduce the product's perceived longevity and reliability, especially in a premium product context where durability and minimalism are key values.

Looking ahead, we anticipate a growing interest in hybrid solutions that retain the simplicity and portability of mechanical operation while selectively integrating electric features that truly enhance the experience. It is likely that user expectations will continue to evolve, especially in the premium market. There may be growing demand for greater convenience, users might expect to simply press one button and let the machine handle the rest.

12.5 Future Work

As this project is a conceptual design exploration, several key areas remain to be addressed in order to develop a production-ready product.

While the main focus has been on the product's external form, interaction design, and user experience, the internal components and mechanical solutions have only been addressed at a preliminary level. In the current 3D CAD model, many internal parts are placeholders and not modeled with exact dimensions. A critical next step involves specifying all required components in detail, including their accurate sizing, placement, and how they connect and interact with other parts of the system. This also includes determining proper mounting solutions and ensuring mechanical and electrical integration is feasible and robust.

Moreover, once the components are finalized, significant programming work will be needed to define and implement the desired behaviors, such as timing, safety sequences, and user feedback, across all systems.

From a mechanical standpoint, the selected components have not yet been dimensioned or validated for the actual loads and pressures experienced during carbonation. For example, it remains to be calculated whether the chosen solenoid can reliably lock the protective cover under carbonation pressure. Additionally, the structural design of the rear column requires further evaluation: it is currently split into two halves, with the internal frame only supporting one side. This solution must be assessed to ensure it can withstand the high internal forces safely and consistently.

Manufacturing considerations have also been beyond the scope of this project. While the general assembly of the product has been explored, production methods for

individual parts, such as part phasing, tolerances, and manufacturing techniques, have not yet been determined. These factors are essential for moving toward mass production and will directly influence feasibility, durability, and cost.

Speaking of cost, it is important to acknowledge that an electrical product, due to its embedded components and systems, will generally require higher production costs compared to a fully mechanical solution. A detailed cost estimate should therefore be conducted to help evaluate whether the increased functionality and user experience justify the added expense.

To transition from concept to production, thorough testing and iterative prototyping will be essential. Physical prototypes that include functional components need to be built and evaluated to validate core mechanisms and user interactions. This would enable refinement of specific features, such as the tactile resistance of the rear column button or the acoustic quality of the solenoid. Such detailed adjustments are only possible once real-world performance and usability data can be collected.

In summary, to bridge the gap between concept and production, several areas remain to be explored:

- Specification and Integration of Internal Components.
- Manufacturing Methodology and DFM (Design for Manufacturing).
- Mechanical Load Analysis and Structural Validation.
- Embedded Programming and Control System Development.
- Prototype Development, Testing and Design Validation.

Chapter 13

Conclusion

This project aimed to improve the interaction design of the Aarke Carbonator Pro, guided by five key objectives. Each has been addressed in the final concept:

1. Improve Intuitiveness: *The redesigned interaction flow uses clear affordances and visual cues to guide the user seamlessly.*
2. Enhance Safety Perception: *Integrated electric features support a secure experience and minimize the potential for user errors.*
3. Streamline User Flow: *Interaction steps were reduced and refined to create a more efficient and ergonomic process.*
4. Clear Feedback: *Visual and auditory cues were introduced to communicate system responses and build user confidence.*
5. Respect Brand Identity: *The final concept preserves Aarke's core values of minimalism, refinement, and timeless design.*

In summary, the project highlights how user-centered design and thoughtful integration of electronic functionality can elevate both usability and brand experience in premium home appliances.

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Appendix A

Observation Material

Observation material for the user study is presented here, including the observation guide and the likert-scale questionnaire.

Observation Guide

The document provides a guide for the conducted user study where both first-time users and experienced users interacted with the Aarke Carbonator Pro. The guide consists of the steps that were included in the study, and questions that were asked to the participants.

Observation Guide

Dokumentation

- Spela in video eller anteckna deras steg och reaktioner.
- Notera tiden det tar att kolsyra en flaska första gången.

Interaktionspunkter

- Fylla på vatten
- Placering av flaska (ställa flaska på rätt plats)
- Aktivering av skydd (trycka på kupolen)
- Kolsyresättning (trycka på knapp)
- Ventilera (dra ner spak)
- Inaktivera skydd (släppa spak)
- Byte av cylinder

Feedback - Frågor

- Placering av flaskan
    - Förstår man vart den ska stå? Varför/Varför inte?
    - Är den lätt att placera? Varför/Varför inte?
    - Känns det säkert och stabilt att placera flaskan? Varför/Varför inte?
- Aktivering av skydd
    - Förstår man interaktionspunkter? Varför/Varför inte?
    - Förstår man vad man ska göra? Varför/Varför inte?
    - Hur är känslan att trycka ner kupolen? Trög? Lätt?
    - Får man feedback på när skyddet har låst fast? Hur? Behöver man mer?
    - Känns skyddet säkert? Varför/Varför inte?
- Kolsyresättning
    - Förstår man interaktionspunkter? Varför/Varför inte?
    - Förstår man vad man ska göra? Varför/Varför inte?
    - Hur är känslan att trycka på knappen? Trög? Lätt?
    - Vad för feedback får man? Hur upplever du den?
- Ventilering
    - Förstår man interaktionspunkter? Varför/Varför inte?
    - Förstår man vad man ska göra? Varför/Varför inte?
    - Hur är känslan att trycka ner spaken? Trög? Lätt?
    - Vad för feedback får man? Hur upplever du den?
- Inaktivering av skydd
    - Förstår man vad man ska göra? Varför/Varför inte?
    - Vad för feedback får man? Hur upplever du den?
- Byte av cylinder
    - Förstår man hur man ska göra? Varför/Varför inte?
    - Hur är upplevelsen?

Oerfaren användare

Syfte & Mål

- Undersöka hur användare interagerar vid första användning.
- Identifiera eventuella problem eller hinder i användarupplevelsen.
- Få insikter om hur tydliga instruktioner och design är.
- Förstå vad som kan förbättras.

Steg 1:

- Filma när de använder pro för första gången utan instruktioner, se hur de interagerar med den. Be dem samtidigt tänka högt och beskriva vad de gör, varför de gör det och vad de uppfattar som svårt eller enkelt.
- Be personen byta cylinder och observera om de förstår hur man ska agera. Be dem samtidigt tänka högt och beskriva vad de gör, varför de gör det och vad de uppfattar som svårt eller enkelt.
- Följdfrågor efter att de fått en liten paus att reflektera
    - Vad var din första tanke när du såg maskinen?
    - Hur tänkte du när du först skulle börja använda den? Vad var dina instinkter?
    - (Baserat på hur de interagerade med den) Förklara steg för steg varför du gjorde som du gjorde

Steg 2:

- Visa hur den ska användas
- Låt dem använda Pro igen (se interaktionspunkter och feedback)
- Vad känns naturligt? Vad kändes svårt? Vad skulle du ändra?
- Vad är känslan när den används?

Erfaren användare

Syfte & Mål

- Förstå vad som fungerar bra och vad som kan förbättras för en erfaren användare.
- Identifiera eventuella små irritationsmoment eller förbättringsområden som nya användare kanske inte märker direkt.
- Undersöka om användarna har utvecklat egna lösningar för att hantera eventuella svagheter i produkten.

Steg 1:

- Filma när de använder maskinen, se hur de interagerar med den. Skiljer det sig något från hur den är tänkt att användas? Hur kommer det sig?

Steg 2:

- Låt dem använda den igen
- Be personen byta cylindern
- Be dem muntligt gå igenom steg för steg hur varje interaktion och feedback känns (se interaktionspunkter och feedback)

Steg 3:

- Vad är känslan när den används?
- Hur ofta använder du pro?
- Var det lätt att lära sig?
- Behöver du tänka när du använder den eller gör du rörelserna per automatik?
- Har det någon gång uppstått något fel eller problem? Har du förstått hur du ska lösa det?
- Finns det något med maskinen som du tycker är irriterande eller kunde fungera smidigare?
- Har du märkt några problem över tid, t.ex. att knappar blir trögare, packningar slits eller att den känns mindre effektiv?
- Har du några tips för att göra det snabbare?
- Har din uppfattning om produkten förändrats sedan du började använda den?

Likert-scale questionnaire

The likert-scale quesionnaire was filled in by the participants after the user test.

Hur var det att förstå hur och var flaskan skulle placeras?

Mycket enkelt 1 2 3 4 5 *Mycket svårt*

Hur var det att förstå hur flaskan skulle omslutas?

Mycket enkelt 1 2 3 4 5 *Mycket svårt*

Hur var det att trycka ner kupolen?

Mycket enkelt 1 2 3 4 5 *Mycket trögt*

Hur upplevdes skyddet för glasflaskan?

Mycket säkert 1 2 3 4 5 *Mycket osäkert*

Hur var det att förstå hur man skulle kolsyresätta?

Mycket enkelt 1 2 3 4 5 *Mycket svårt*

Hur kändes det att trycka på knappen för att kolsyresätta?

Mycket enkelt 1 2 3 4 5 *Mycket trögt*

Hur var feedbacken vid kolsyresättning?

Mycket tydlig 1 2 3 4 5 *Mycket otydlig*

Hur lätt var det att förstå hur man skulle släppa på trycket och ventilera?

Mycket enkelt 1 2 3 4 5 *Mycket svårt*

Hur var feedbacken vid ventilering?

Mycket tydlig 1 2 3 4 5 *Mycket otydlig*

Hur lätt var det att förstå hur man skulle öppna på skyddet?

Mycket enkelt 1 2 3 4 5 *Mycket svårt*

Hur var det att trycka ner spaken?

Mycket enkelt 1 2 3 4 5 *Mycket trögt*

Hur var det att förstå hur man skulle byta ut cylinder?

Mycket enkelt 1 2 3 4 5 *Mycket svårt*

Hur upplevdes hela användningen av kolsyremaskinen?

Mycket intuitiv 1 2 3 4 5 *Svår att förstå*

Skulle du rekommendera kolsyremaskinen vidare?

Mycket troligt 1 2 3 4 5 *Inte alls troligt*

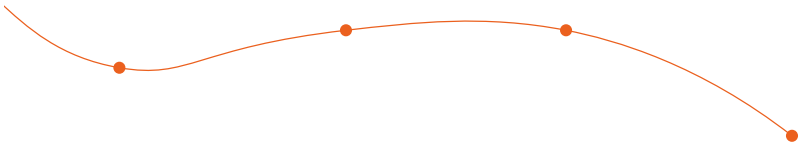
Appendix B

Customer Journey Map

After the user study, a customer journey map was conducted to go through each step of the process and elaborate on needs and pain points. This led to opportunities for suggestions on how to improve the carbonation process.

Steps	Preparation	Loading machine	Activate cover	Carbonation
Actions	• Get the bottle • Take forward carbonator • Fill bottle with water until the filling line	• Place the bottle in the designated area in the base	• Press down the cover to enclosure the bottle until it klicks and fastens	• Carbonate water by pressing down button until a buzzing sound • Press more to get more bubbles
Needs & Pain Points	• Clear directions of how much water to fill • Place carbonator whereever • Aesthetically pleasing to always have on counter	• Clear where to place the bottle • Easy and fast to place bottle using one hand • Feeling that the bottle is secure and fits well	• Clear how to perform the action • Not too much force required • Feel that the cover is safe and secured	• Not too much force to press button • Clearly shown where and how to carbonate • Feedback during carbonation and when done • Personalize amount of CO2
Feelings				
Opportunities		• Easier to put in bottle, not having to tilt it	• Make it easier to cover the bottle, less force required, easier to understand how to	• Feedback in button during carbonation • Decide how much to carbonate • Clearly show carbonation level

Ventilation	Raise cover	After use	Replace cylinder
• Press down the handle on the side slowly to release the pressure and ventilate	• Raise cover by pressing down the handle all the way • Cover opens by itself • Take out bottle with sparkling water	• Clean machine • Wash bottle	• Lay carbonator on side • Unscrew cylinder to remove it • Remove plastic cover and stopper on the new cylinder
• Clearly shown how to ventilate • Feedback in when pressure is released and when ventilation is done	• Needs to feel safe to open the cover	• Easy to clean machine • Not getting stains easily • Being able to wash bottle in dishwasher	• Not too heavy to tilt the machine • To much force needed to screw the cylinder • Too tight for the hand to screw • Know when the cylinder is attached



• More intuitive motion in the handle • Feedback in interaction point	• More intuitive motion for opening the cover		• Not having to tilt the machine • Clearly show how much gas i left or when you need to replace cylinder

Appendix C

Requirements List

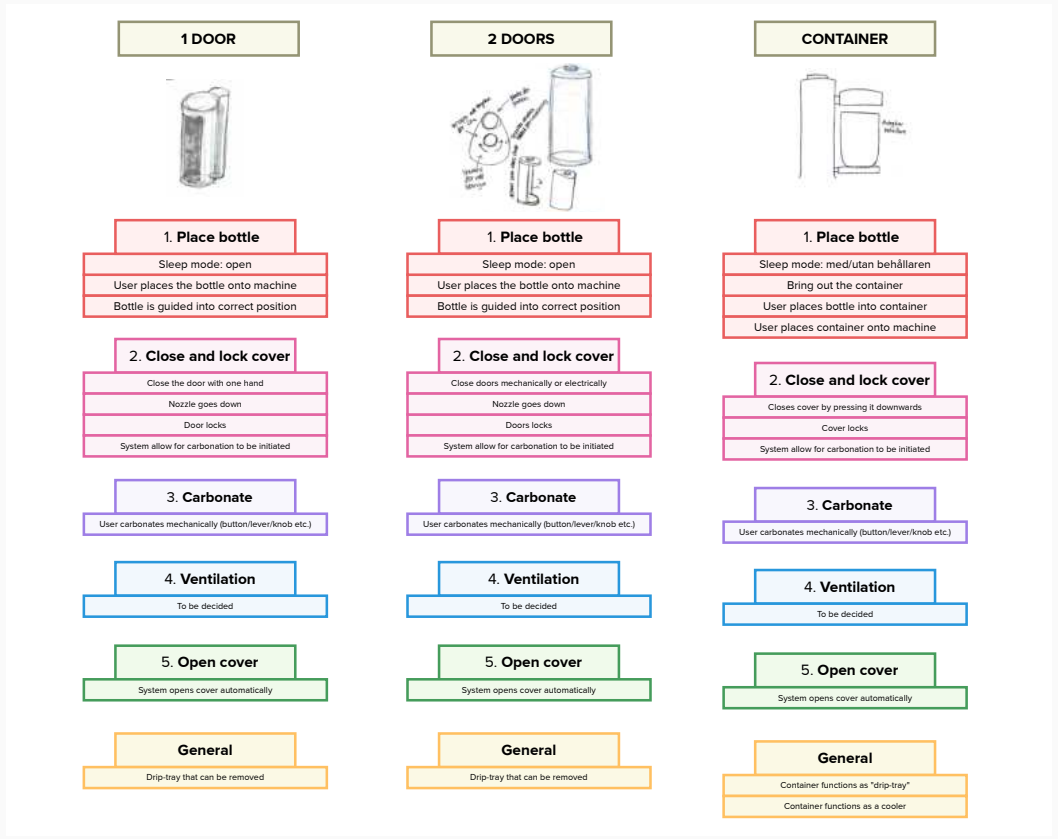
The requirements list was put together based on the insights and regarding the electrical part of the product. The requirements were divided into essential and desirable.

Requirements	Essential/ Desirable	Comment
Compatible with standard CO2 cylinder	E	425 g CO2 cylinder
Rechargeable	E	Cordless functionality, charge with USB-C
Battery capacity	D	50 uses per charge
Electrical safety	E	IPX4-standard
Length of charging cable	D	Minimum 90 cm
Not carbonate unless bottle is encapsulated	E	Safety
Collect spilled water	D	
Sustainable and food-grade materials	E	
Quick usage process	D	Carbonate a bottle within maximum 30 sec
Enable one-handed use	E	
Height	D	Maximum height of 50 cm
Weight	D	Maximum weight of 4 kg
Fit in Aarkes product portfolio	E	
Express safety	D	
Express robustness	D	
Not be able to carbonate empty bottle	E	Safety
Protective cover must withstand high pressure	E	The pressure generated when carbonating an empty bottle
Be able to use without reading manual	D	Intuitveness

Appendix D

Flow Charts of Bottle Encapsulation Concepts

For the three bottle encapsulation concepts, flow charts were made to analyze how the use steps would look like for each concept. These flow charts were used to compare the three concepts during the concept selection phase.


1 DOOR
1. Place bottle
Sleep mode: open
User places the bottle onto machine
Bottle is guided into correct position
2. Close and lock cover
Close the door with one hand
Nozzle goes down
Door locks
System allow for carbonation to be initiated
3. Carbonate
User carbonates mechanically (button/lever/knob etc.)
4. Ventilation
To be decided
5. Open cover
System opens cover automatically
General
Drip-tray that can be removed
2 DOORS
1. Place bottle
Sleep mode: open
User places the bottle onto machine
Bottle is guided into correct position
2. Close and lock cover
Close doors mechanically or electrically
Nozzle goes down
Doors locks
System allow for carbonation to be initiated
3. Carbonate
User carbonates mechanically (button/lever/knob etc.)
4. Ventilation
To be decided
5. Open cover
System opens cover automatically
General
Drip-tray that can be removed
CONTAINER
1. Place bottle
Sleep mode: med/utan behållaren
Bring out the container
User places bottle into container
User places container onto machine
2. Close and lock cover
Closes cover by pressing it downwards
Cover locks
System allow for carbonation to be initiated
3. Carbonate
User carbonates mechanically (button/lever/knob etc.)
4. Ventilation
To be decided
5. Open cover
System opens cover automatically
General
Container functions as "drip-tray"
Container functions as a cooler

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